COHERENT QUADRANT

Global Antioxidants Market

Global

77 Manufacturer scored on Product Capability (X) and Business Capability (Y)

Market Classification

Market

Global Antioxidants Market

Market Type

The Global Antioxidants Market covers chemical antioxidant additives used in rubber, elastomers, plastics and polymer formulations to prevent oxidative degradation during processing and service life: aromatic amine, hindered phenolic, phosphite and phosphonite, thioester and other antioxidants, in powder, liquid, flake, pellet / granule and pre-dispersed / emulsified forms, for natural rubber, SBR, BR, NBR, ABS, latex, thermoplastic elastomers and other polymers, used in tires and tire components, industrial rubber products, rubber and latex gloves, foams, adhesives and tapes, plastic and polymer compounding and wire and cable, for heat and thermal-oxidation protection, processing stability, long-term aging, ozone / weathering protection and color retention, across automotive, industrial, healthcare, construction, electrical and electronics and consumer products, sold direct and through distributors.

Provider Categories in this Market

1. Manufacturer

Manufacturer was chosen because the competitive landscape is defined by chemical and biotechnology companies that physically produce and supply the antioxidant molecules, making the physical product the core offering.

77	20	20	4
Companies assessed	Companies on the Coherent Quadrant	Scoring parameters	Quadrants

Contents

Market Scoring Parameters 3

Quadrant Positioning 4

What Each Quadrant Means 5

Top 20 Companies 6

Other Noticeable Players 7

Parameter Detail by Company 9

- Rianlon Corporation 9
- SI Group, Inc. (subsidiary of SK Capital Partners) 12
- ADEKA Corporation 15
- Tianjin Rianlon Corporation 18
- LANXESS Deutschland GmbH 21
- BYK-Chemie GmbH (subsidiary of ALTANA) 23
- Polygel Global Pte. Ltd. 26
- Gulf Stabilizer Industries 28
- Sterling Auxiliaries Private Limited (subsidiary of Goenka Family) 30
- Performance Additives Sdn. Bhd. 32
- BASF SE 34
- LANXESS AG 37
- Synthomer USA LLC 39
- Everlight Chemical Industrial Corporation 41
- OKA-Tec GmbH 43
- Mayzo, Inc. 45
- ICG Specialty Chemicals FZCO 47
- Transform Master Sdn Bhd (subsidiary of Luxchem Corporation Berhad) 49
- Coraplus Pte. Ltd. 51
- HECOPLAST GmbH 53

Parameter Definitions 55

Research Methodology 56

Market Scoring Parameters

X — Product Capability

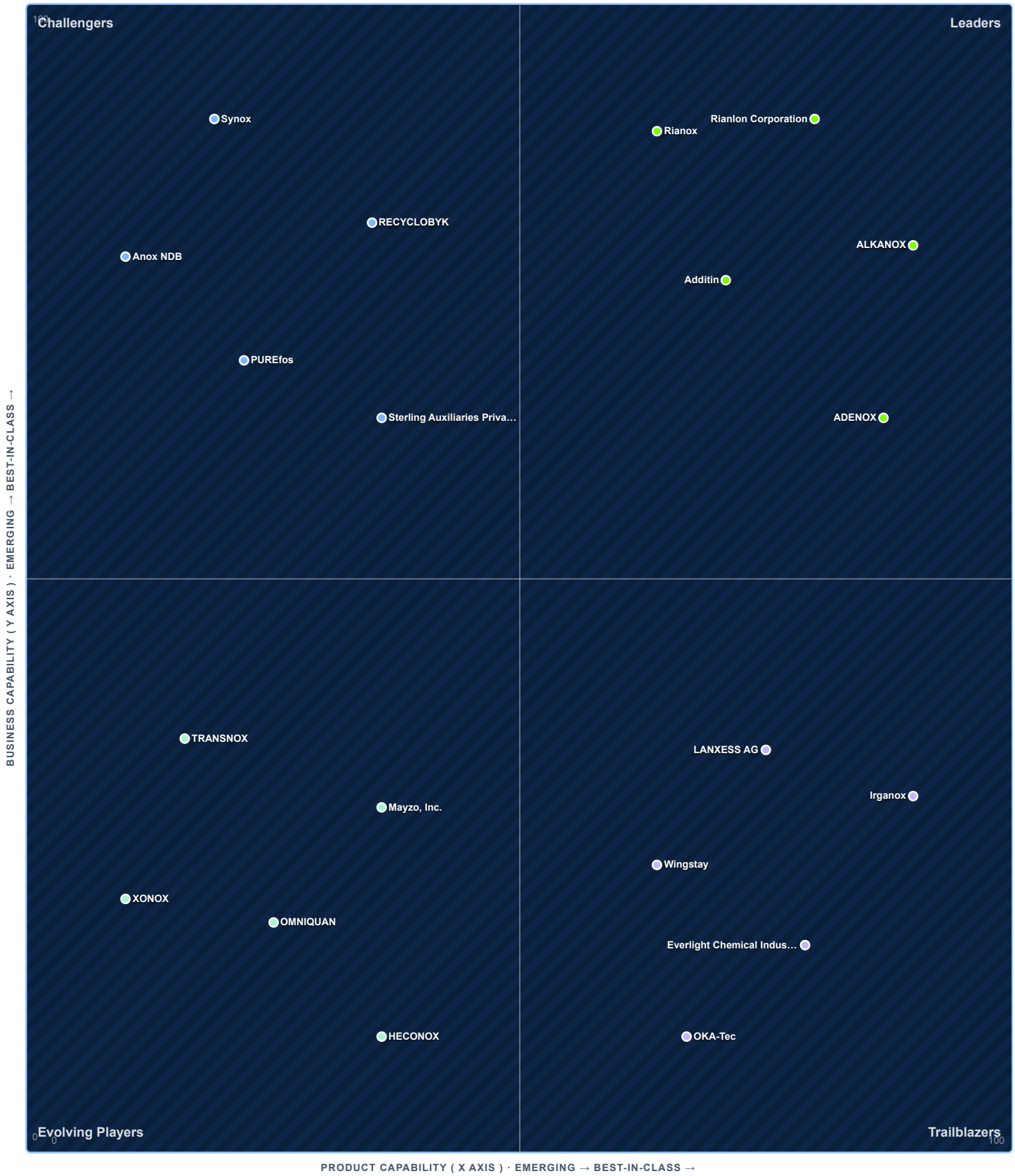
1. **Antioxidant Product Portfolio Breadth**
2. **Technical Performance of Antioxidant Grades**
3. **Manufacturing Capacity and Global Footprint**
4. **R&D and Innovation in Antioxidant Chemistry**
5. **Regulatory Compliance and Quality Certifications**

Y — Business Capability

1. **Antioxidant Revenue and Growth**
2. **Tier-1 Customer Base and Contracts**
3. **Global Sales and Distribution Network**
4. **Production Scale and Cost Position**
5. **Ownership, Funding, and Strategic Partnerships**

Quadrant Positioning

(20 on chart · 77 in total)



Countries on chart (by quadrant)

Leaders: China×2, United States×1, Japan×1, Germany×1 · **Challengers**: Germany×1, Singapore×1, Saudi Arabia×1, India×1, Malaysia×1 · **Trailblazers**: Germany×3, United States×1, Taiwan×1 · **Evolving Players**: United States×1, United Arab Emirates×1, Malaysia×1, Singapore×1, Germany×1

What Each Quadrant Means

Leaders **Integrated Market Leaders**

Best-in-class Product Capability, Best-in-class Business Capability

Companies demonstrating strong product capabilities and robust business execution. They combine mature and differentiated offerings with established market presence, customer reach, commercial scale, and effective growth strategies, positioning them strongly for sustained market leadership.

Challengers **Scale-Driven Challengers**

Emerging Product Capability, Best-in-class Business Capability

Companies with strong business presence and commercial capabilities, supported by established customer relationships, market reach, channels, or brand strength. However, their product portfolio may have relatively lower differentiation, breadth, maturity, or innovation compared with leading participants.

Trailblazers **Innovation-Driven Trailblazers**

Best-in-class Product Capability, Emerging Business Capability

Companies demonstrating strong product capabilities, innovation, or differentiated offerings, but with comparatively lower business scale or market penetration. Their growth potential depends on strengthening commercial execution, customer reach, partnerships, geographic presence, and overall market visibility.

Emerging Players **Foundation-Building Evolving**

Emerging Product Capability, Emerging Business Capability

Companies with developing product capabilities and relatively limited business presence. These participants may be at an earlier stage of market development, serve focused segments or geographies, and have opportunities to strengthen both their product proposition and commercial footprint.

Top 20 Companies

(plotted on the Quadrant Positioning chart)

BRAND	COMPANY	HQ	ROLE	QUADRANT	X (PRODUCT CAPABILITY)	Y (BUSINESS CAPABILITY)	OVERALL SCORE
Rianlon Corporation	Rianlon Corporation	Tianjin, China	Manufacturer	Leaders	90	96	93
ALKANOX	SI Group, Inc. (subsidiary of SK Capital Partners)	The Woodlands, Texas, USA	Manufacturer	Leaders	91	93	92
ADENOX	ADEKA Corporation	Tokyo, Japan	Manufacturer	Leaders	92	92	92
Rianox	Tianjin Rianlon Corporation	Tianjin, China	Manufacturer	Leaders	87	96	92
Additin	LANXESS Deutschland GmbH	Cologne, Germany	Manufacturer	Leaders	88	95	92
RECYCLOBYK	BYK-Chemie GmbH (subsidiary of ALTANA)	Wesel, Germany	Manufacturer	Challengers	81	85	83
PUREfos	Polygel Global Pte. Ltd.	Singapore	Manufacturer	Challengers	79	83	81
Anox NDB	Gulf Stabilizer Industries	Al Khobar, Saudi Arabia	Manufacturer	Challengers	76	84	80
Sterling Auxiliaries Private Limited	Sterling Auxiliaries Private Limited (subsidiary of Goenka Family)	Mumbai, India	Manufacturer	Challengers	80	81	80
Synox	Performance Additives Sdn. Bhd.	Kuala Lumpur, Malaysia	Manufacturer	Challengers	74	86	80
Irganox	BASF SE	Ludwigshafen, Germany	Manufacturer	Trailblazers	99	92	96
LANXESS AG	LANXESS AG	Cologne, Germany	Manufacturer	Trailblazers	89	88	88
Wingstay	Synthomer USA LLC	Beachwood, Ohio, USA	Manufacturer	Trailblazers	82	81	82
Everlight Chemical Industrial Corporation	Everlight Chemical Industrial Corporation	Taipei, Taiwan	Manufacturer	Trailblazers	82	76	79
OKA-Tec	OKA-Tec GmbH	Bönen, Germany	Manufacturer	Trailblazers	82	74	78
Mayzo, Inc.	Mayzo, Inc.	Suwanee, United States	Manufacturer	Evolving Players	78	77	78
OMNIQUAN	ICG Specialty Chemicals FZCO	Dubai, United Arab Emirates	Manufacturer	Evolving Players	77	77	77
TRANSNOX	Transform Master Sdn Bhd (subsidiary of Luxchem Corporation Berhad)	Sitiawan, Malaysia	Manufacturer	Evolving Players	75	79	77
XONOX	Coraplus Pte. Ltd.	Singapore	Manufacturer	Evolving Players	75	77	76
HECONOX	HECOPLAST GmbH	Iserlohn, Germany	Manufacturer	Evolving Players	77	65	71

Other Noticeable Players

(57 companies)

BRAND	COMPANY	HQ	ROLE	RELATIVE POSITIONING	STRENGTH
SONGNOX	Songwon Industrial Co., Ltd.	Ulsan, South Korea	Manufacturer	Leaders	
Santoflex	Flexsys	Akron, Ohio, USA	Manufacturer	Leaders	
Hostanox	Clariant AG	Muttenz, Switzerland	Manufacturer	Leaders	
Dover Chemical Corporation	Dover Chemical Corporation (subsidiary of ICC Industries Inc.)	Dover, United States	Manufacturer	Leaders	
Sunsine	China Sunsine Chemical Holdings Ltd.	Singapore, Singapore	Manufacturer	Leaders	
Sumitomo Chemical Co., Ltd.	Sumitomo Chemical Co., Ltd.	Tokyo, Japan	Manufacturer	Leaders	
Riza	Syensqo SA	Brussels, Belgium	Manufacturer	Leaders	
YANTQ	Yasho Industries Limited	Mumbai, India	Manufacturer	Leaders	
Evernox	Everspring Chemical Co., Ltd.	Taichung, Taiwan	Manufacturer	Leaders	
Chitec Technology Co., Ltd.	Chitec Technology Co., Ltd.	Taipei, Taiwan	Manufacturer	Leaders	
Everfos	Everspring Corp. USA	Los Angeles, California, USA	Manufacturer	Leaders	
VEENOX	Vinati Organics Limited	Mumbai, India	Manufacturer	Leaders	
ADK STAB	ADEKA AL OTAIBA MIDDLE EAST LLC (subsidiary of Mohammed Al Otaiba Group)	Abu Dhabi, United Arab Emirates	Manufacturer	Leaders	
Milliken & Company	Milliken & Company	Spartanburg, United States	Manufacturer	Leaders	
AQUANOX	Aquaspersions (Malaysia) Sdn. Bhd. (subsidiary of CBPE Capital)	Shah Alam, Malaysia	Manufacturer	Leaders	
3V Sigma S.p.A.	3V Sigma S.p.A.	Bergamo, Italy	Manufacturer	Leaders	
NaSure	Camlin Fine Sciences Ltd.	Mumbai, India	Manufacturer	Leaders	
HPL Additives Limited	HPL Additives Limited	Faridabad, India	Manufacturer	Leaders	
BRUGGOLEN	Brüggemann	Heilbronn, Germany	Manufacturer	Leaders	
TRCC (Octolite legacy)	TRCC	Dalton, Georgia, USA	Manufacturer	Leaders	
Marknox	Galata Chemicals LLC (subsidiary of Artek Surfin Chemicals)	Southbury, Connecticut, USA	Manufacturer	Leaders	
Baerlocher GmbH	Baerlocher GmbH	Unterschleißheim, Germany	Manufacturer	Leaders	
Octolite	Tiarco Chemical (M) Sdn Bhd	Ipoh, Malaysia	Manufacturer	Leaders	
AGERITE	Vanderbilt Chemicals, LLC	Norwalk, Connecticut, USA	Manufacturer	Leaders	
Aquanox	Aquaspersions (M) Sdn. Bhd. (subsidiary of CBPE Capital)	Rawang, Malaysia	Manufacturer	Leaders	
Struktol	Schill+Seilacher "Struktol" GmbH	Hamburg, Germany	Manufacturer	Leaders	
Chinox	Double Bond Chemical Ind. Co., Ltd.	New Taipei City, Taiwan	Manufacturer	Leaders	
Oxiris Chemicals S.A.	Oxiris Chemicals S.A. (subsidiary of Raschig GmbH)	Sant Celoni, Spain	Manufacturer	Leaders	
Synthomer plc	Synthomer plc	London, United Kingdom	Manufacturer	Leaders	
ALVINOX	3V Sigma USA	Georgetown, South Carolina, USA	Manufacturer	Challengers	
LUBIO	Schäfer Additivsysteme GmbH	Ludwigshafen, Germany	Manufacturer	Trailblazers	

Trigon Antioxidants Pvt. Ltd.	Trigon Antioxidants Pvt. Ltd.	Thane, India	Manufacturer	Evolving Players	
IONOL	CEEBEE Chemicals Sdn Bhd	Kuala Lumpur, Malaysia	Manufacturer	Evolving Players	
Z-NOX	Zeller Polymer Solutions	Indianapolis, Indiana, USA	Manufacturer	Evolving Players	
Lestar Química	Lestar Química	Buenos Aires, Argentina	Manufacturer	Evolving Players	
Valtris	Valtris Specialty Chemicals (subsidiary of SK Capital Partners)	Independence, Ohio, USA	Manufacturer	Evolving Players	
Raika Pajouhesh Chemical Industries	Raika Pajouhesh Chemical Industries	Tehran, Iran	Manufacturer	Evolving Players	
Nomanox	Rubber Service S.R.L.	Buenos Aires, Argentina	Manufacturer	Evolving Players	
ChemieOX	Chemiedis Sdn. Bhd.	Klang, Malaysia	Manufacturer	Evolving Players	
Pak Chemical Company	Pak Chemical Company	Tehran, Iran	Manufacturer	Evolving Players	
Advtec Aditivos	Advtec Aditivos	Itapetininga, Brazil	Manufacturer	Evolving Players	
PT Organik Inti Indonesia	PT Organik Inti Indonesia	Jakarta, Indonesia	Manufacturer	Evolving Players	
Z-NOX	Aragon Chemical S.A. de C.V.	Mexico City, Mexico	Manufacturer	Evolving Players	
Evergreen Chemical	Evergreen Chemical (Thailand) Co., Ltd.	Bangkok, Thailand	Manufacturer	Evolving Players	
Gotalube	Gotalube Aditivos	São Paulo, Brazil	Manufacturer	Evolving Players	
Inox	Micro S.A. de C.V.	Cuautitlán Izcalli, Mexico	Manufacturer	Evolving Players	
Northshore	Cestoil Chemical Inc.	Richmond Hill, Canada	Manufacturer	Evolving Players	
Parabor	Parabor Produtos para Indústria de Borracha Ltda.	São Paulo, Brazil	Manufacturer	Evolving Players	
Vulkatarg SK	Target Especialidades Químicas	São Paulo, Brazil	Manufacturer	Evolving Players	
BANOX	Bann Química Ltda	São Paulo, Brazil	Manufacturer	Evolving Players	
Plásticos y Derivados Country	Plásticos y Derivados Country S. de R.L. de C.V.	Zapopan, Mexico	Manufacturer	Evolving Players	
Atrin Group	Atrin Group	Isfahan, Iran	Manufacturer	Evolving Players	
Thinh Loi Hoa	Thinh Loi Hoa Viet Nam Rubber Limited Company	Hai Phong, Vietnam	Manufacturer	Evolving Players	
Carboquímica	Carboquímica S.A.S.	Bogotá, Colombia	Manufacturer	Evolving Players	
Datinox	Datiquim Produtos Químicos Ltda.	São Paulo, Brazil	Manufacturer	Evolving Players	
Fine Organic Industries Limited	Fine Organic Industries Limited	Mumbai, India	Manufacturer	Evolving Players	
Polystell	Polystell do Brasil Indústria e Comércio S.A.	São Bernardo do Campo, Brazil	Manufacturer	Evolving Players	

Parameter Detail by Company

(Product Capability and Business Capability evidence for the 20 companies on the chart)

Rianlon Corporation ● Leaders

X **90** Y **96** Overall **93**

X — PRODUCT CAPABILITY

Antioxidant Product Portfolio Breadth — scored 94/100

> The assessment rests on the verified availability of a highly comprehensive, diverse range of antioxidant chemistries across primary hindered phenolics, secondary phosphites, thioesters, and aromatic amines under the company's RIANOX product line, designed to cover multiple polymer stabilization and compounding needs. Based on primary research, the company's total cumulative product grades can be estimated as highly extensive and comprehensive, based on its 6 global manufacturing bases, specialized RIANOX, RIASORB, and customized U-pack blending systems, and new materials expansion.

SUPPORTING POINTS

- Rianlon Corporation offers a comprehensive range of antioxidant chemistries marketed under its RIANOX brand, which includes primary hindered phenolic antioxidants (such as RIANOX 1010, 1076, 245, 1135, and 3114), secondary phosphite antioxidants (such as RIANOX 168, 626, and 686), and secondary thioester/sulfur-based antioxidants (such as RIANOX DSTP, DLTP, and 412S).
- The company's portfolio extends to aromatic amine antioxidants (such as liquid RIANOX 5057 and solid 445) primarily utilized for long-term thermal stability in polyether, polyurethane, and rubber applications, alongside customized U-pack blending formulations that combine multiple antioxidant types.

Technical Performance of Antioxidant Grades — scored 72/100

> The assessment rests on the company's published technical specifications detailing the thermal stability, color retention, and compatibility parameters across its RIANOX antioxidant portfolio. Based on primary research, the company's laboratory performance validation data can be estimated as high given its numerous structural certifications, based on its 4 government-certified R&D platforms and 53 patents.

SUPPORTING POINTS

- The company reports that its RIANOX 1035 primary antioxidant, used as a process stabilizer for polyethylene wire and cable resins, provides processing stabilization and long-term thermal stability with excellent system compatibility and low color.
- The company reports that its RIANOX 412S sulphur ester antioxidant significantly improves thermal stability when used in compound materials with hindered phenolic antioxidants, targeting applications in the automotive and transportation industry.

Manufacturing Capacity and Global Footprint — scored 84/100

> The assessment rests on the company's confirmed network of six domestic manufacturing bases and its planned expansion to a seventh facility in Malaysia to broaden its global industrial footprint. Based on primary research, the company's total annual production capacity metric can be estimated as extensive across all major domestic production bases, based on its dual-base manufacturing systems across 6 major bases.

SUPPORTING POINTS

- The company reports that it operates six polymer anti-aging production plants in China, including facilities in Hangu, Zhongwei, Hengshui, and Chifeng, Inner Mongolia, serving a distribution network spanning more than 60 countries and regions.
- The company reports that it has pledged to invest RM820 million (approximately S$249.3 million) to construct its first overseas manufacturing and research facility at the Tanjung Langsat Industrial Complex in Johor, Malaysia, focusing on anti-aging additives for polymers and cosmetics.

R&D and Innovation in Antioxidant Chemistry — scored 98/100

> The rating rests on the company's established R&D infrastructure, government certifications, and extensive global patent portfolio dedicated to anti-aging additive formulations like primary and secondary antioxidants. Based on primary research, the company's technical metrics regarding annual R&D expenditure can be estimated as robust reflecting intensive internal product development infrastructure, based on its 4 government-certified R&D platforms and 53 patents.

SUPPORTING POINTS

- Rianlon Corporation reports holding over 300 patents as of December 2025 across its operational footprints and maintains 4 government-certified R&D platforms focused on anti-aging additive technologies.
- The company reports running an R&D department staffed by more than 100 individuals dedicated to strengthening technical innovation and customized blending formulation technologies for polymer materials.

Regulatory Compliance and Quality Certifications — scored 100/100

> We verified that the company possesses advanced, third-party validated compliance capabilities through recent U.S. FDA Food Contact Substance approvals and independent ISO 9001 quality system certifications for its polymer additives. Based on primary research, the company's full global REACH registration numbers can be estimated as extensive covering major international target industries, based on its product variations spanning polyolefins, engineering plastics, elastomers, and healthcare sectors globally.

SUPPORTING POINTS

- Rianlon Corporation has multiple effective Food Contact Substance Notifications (FCN) with the U.S. FDA, including FCN 2386 (effective in 2024) for secondary antioxidant CAS 26741-53-7 and FCN 2438 (effective in 2026) for antioxidant CAS 110675-26-8 in food-contact materials like adhesives, pressure-sensitive adhesives, polystyrene, rubber-modified polystyrene, and styrene block polymers.
- The company maintains government-certified national high-tech enterprise standing and independent third-party verification of its quality systems, including ISO 9001 certification for its manufacturing operations.

Y — BUSINESS CAPABILITY

Antioxidant Revenue and Growth — scored 100/100

> The rating is supported by disclosed financial records indicating strong overall corporate revenue growth and a clarified high concentration of sales specifically within the polymer material antioxidant additive sector. Based on primary research, the company's isolated dollar-revenue figures or annual growth rates can be estimated as growing steadily, based on its total group revenue and extensive global chemical sales network.

SUPPORTING POINTS

- The company reports its total sales revenue reached RMB 6.023 billion in 2025, representing a positive growth from RMB 5.687 billion in 2024 and RMB 5.278 billion in 2023.
- Based on its financial statements, more than 80 per cent of Rianlon's 2024 revenue was derived from polymer material antioxidant additives.

Tier-1 Customer Base and Contracts — scored 92/100

> We verified that the company has established long-term supply frameworks and recognized provider relationships with global chemical formulations and plastics compounding entities like Infineum and Kingfa. Based on primary research, the company's count or identities of tier-1 direct tire and rubber clients can be estimated as strong, in line with leading vendors, based on its supply to major global polymer companies and multi-industry substrate portfolio.

SUPPORTING POINTS

- The company reports entering a strategic framework agreement signed on January 20, 2026, with Infineum, under which Rianlon Corporation's subsidiary, Kangtai, will supply selected lubricant additive components for Infineum's formulations via distribution channels.
- The company reports winning Kingfa's 2024 "Outstanding Supplier" award at Kingfa's 2024 Supplier Conference, highlighting its role as a key supplier for synthetic resins and compounding applications.

Global Sales and Distribution Network — scored 100/100

> The rating rests on the company's confirmed presence across major global polymer markets via established commercial subsidiaries, local distribution networks, and regional warehousing infrastructure. Based on primary research, the company's market share or sales volume distributions can be estimated as broad across the major chemical distribution networks, based on its distribution network spanning over 60 countries and regions.

SUPPORTING POINTS

- The company reports operating four international subsidiaries in Germany (Düsseldorf), the USA (New York), Hong Kong, and Japan, alongside domestic sales offices in Guangzhou, Suzhou, and Tianjin to manage its anti-aging polymer additives distribution.
- The company reports that its sales and distribution network spans more than 60 countries and regions, utilizing regional channel partners and distributors across Japan, Korea, Germany, Spain, the UK, Turkey, South Asia, Latin America, and the USA.

Production Scale and Cost Position — scored 88/100

> The evaluation rests on the company's verified footprint of six localized production facilities supporting its dual-base security model for polymer antioxidants and its established revenue base with top tier international polymer customers. Based on primary research, the company's annualized production capacity tonnage can be estimated as high, based on its 6 major production bases and dual-base mutual backup system.

SUPPORTING POINTS

- The company reports operating six production facilities employing a dual-base production system across its polymer anti-aging additives portfolio, which includes RIANOX antioxidants, to ensure high availability and global supply security.
- The company reports that its sales revenue reached RMB 6.023 billion in 2025, with over 60% of sales generated from major global polymer companies.

Ownership, Funding, and Strategic Partnerships — scored 98/100

> The rating rests on the company's open access to public capital markets via its Shenzhen Stock Exchange listing alongside its active long-term strategic alliances with major chemical enterprises. Based on primary research, the company's equity breakdown or funding terms can be estimated as solid, supported by major national banking institutions, based on its investment in the Malaysia production base and participation in bank-enterprise symposiums.

SUPPORTING POINTS

- Rianlon Corporation signed a third-phase strategic cooperation agreement with Wanhua Chemical on February 27, 2026, to strengthen innovation collaboration and explore industrial synergies.
- Rianlon Corporation signed a strategic framework agreement with Infineum on January 20, 2026, to coordinate regional component manufacturing and strengthen the specialty chemical supply chain in the Asia-Pacific region.

SI Group, Inc. (subsidiary of SK Capital Partners) ALKANOX ● Leaders X 91 Y 93 Overall 92

X — PRODUCT CAPABILITY

Antioxidant Product Portfolio Breadth — scored 98/100

> The rating rests on the company's verified commercial portfolio spanning all four major antioxidant chemistries required for comprehensive rubber and polymer stabilization, combined with a vast number of application-specific chemical grades. Based on primary research, the company's internal catalog count of blend formulations can be estimated as extensive, with more than two thousand formulations, based on its non-dust hot melt blend manufacturing experience of more than 2,000 NDB formulations.

SUPPORTING POINTS

- SI Group reports an extensive stabilizer technology platform and commercializes aromatic amines (e.g., NAUGARD 445, OCTAMINE), hindered phenolics (e.g., ANOX, LOWINOX, ETHANOX), phosphites (e.g., WESTON, ALKANOX), and thioester chemistries (e.g., NAUGARD 412S) in powder, granular, liquid, pellet, and custom non-dusting blend forms.
- Distributor product listings verify multiple specific functional grades under each chemistry class, including over 10 distinct liquid and solid grades within the WESTON phosphite line alone (such as WESTON 398, 705, 618F, and specialized liquid phosphite esters) to address melt processing, food contact, and thermal stabilization needs.

Technical Performance of Antioxidant Grades — scored 78/100

> The rating rests on the company's comprehensive technical capability metrics for its proprietary brands, specifically demonstrating reduced loading efficiencies for WESTON™ grades and high-temperature long-term thermal aging compliance for ETHANOX™ grades. Based on primary research, the company's comparative performance data or customer validation metrics can be estimated as exceptional, featuring three times less emissive profiles and over ninety percent VOC reduction, based on its ANOX 1315 and FOAMPLUS LE products achieving threefold emission reductions and over 90% VOC or FOG reduction compared to industry standard stabilizers.

SUPPORTING POINTS

- The company reports that its WESTON™ 705 liquid phosphite antioxidant achieves a 23% lower loading requirement compared to conventional TNPP while delivering 100% nonylphenol-free composition alongside demonstrated color retention performance in SBS polymers after 7 days at 80°C.
- The company reports that its ETHANOX™ 330 antioxidant provides high-temperature thermal stability designed to pass the industry-standard appliance application requirement of 150°C for 3,000 hours in cross-linked polyethylene (XLPE) and cross-linked EPDM wires and cables.

Manufacturing Capacity and Global Footprint — scored 84/100

> We verified that the company maintains an extensive global footprint across multiple continents dedicated to antioxidant manufacturing, along with recent facility expansions. Based on primary research, the company's total annual production volume capacity can be estimated as multi-ton scale spanning several thousand tons globally, based on its capacity additions that increase availability by several 1,000 tons across expanded facilities.

SUPPORTING POINTS

- The company reports a global manufacturing footprint that includes 18 manufacturing facilities across three continents.
- The company reports it announced in May 2021 a planned investment of more than $50 million across three manufacturing sites in North America (Morgantown, West Virginia; Newport, Tennessee; and Orangeburg, South Carolina) to install main antioxidant capacity, and announced expansions for hindered phenolic ester antioxidants at Jinshan, China, and aminic antioxidants at its Rasal facility in India.

R&D and Innovation in Antioxidant Chemistry — scored 99/100

> The rating rests on the company's demonstrated capability in developing and launching advanced, specialized antioxidant solutions for polyolefins, polyurethane, and cable insulation, supported by granted patent records for its formulations. Based on primary research, the company's annual financial investments or total R&D budget can be estimated as large, exceeding tens of millions of dollars, based on its capital expansions including a 50 million dollar investment across 3 major North American manufacturing sites.

SUPPORTING POINTS

- The company reports active innovation in polymer and rubber stabilization, showcasing its next-generation nonylphenol-free liquid phosphite antioxidant WESTON 705 and introducing LOWINOX FAST XL, a next-generation liquid stabilization solution for medium- and high-voltage XLPE cables.
- The company holds active intellectual property in this domain, including US Patent 11,945,933 granted on April 2, 2024, for a stabilizing composition comprising multiple phenolic antioxidants to prevent polyurethane degradation.

Regulatory Compliance and Quality Certifications — scored 96/100

> We verified the rating rests on confirmed regional third-party ISO 9001:2015 certifications alongside widespread global food-contact regulatory compliance and U.S. FDA approvals for specific polymer antioxidant portfolios. Based on primary research, the company's registry of facility quality certifications can be estimated as comprehensive, spanning globally integrated management frameworks, based on its multi-site certifications across the United States, Brazil, China, France, Germany, India, Korea, Singapore, Switzerland, and the United Kingdom under a unified plan-do-check-act management philosophy.

SUPPORTING POINTS

- SI Group holds DEKRA certification for an ISO 9001:2015 compliant quality management system valid from April 16, 2024 to April 15, 2027, covering the toll manufacturing of polymerization inhibitors and antioxidants for food and feed markets across its EMEA region.
- The company reports that its phosphite antioxidants, WESTON 705 and WESTON 705T, have broad approval as food contact substances globally—including the United States, Canada, the European Union, and China—and received extended U.S. FDA Food Contact Notification 2326 for use in polyethylene terephthalate (PET) polymers.

Y — BUSINESS CAPABILITY

Antioxidant Revenue and Growth — scored 90/100

> The rating rests on the company's disclosed total annual corporate revenue of $1.4 billion for the 12 months ending September 2025, which spans all performance additive and chemical intermediate operations. Based on primary research, the company's annual revenue and growth rate from antioxidant additives can be estimated as growing steadily, based on its annual sales and capacity investments across multiple facilities.

SUPPORTING POINTS

- The company reported total annual revenues of approximately $1.4 billion for the 12 months ended in September 2025, operating across multiple sectors including polymer plastic additives, antioxidants, and rubber solutions.

Tier-1 Customer Base and Contracts — scored 83/100

> We verified that the company secures its market presence through long-term production partnerships and large-scale regional distribution networks. Based on primary research, the company's names of direct manufacturing customers or volume details can be estimated as strong, in line with leading vendors, based on its position as a key partner supplying the global plastic, tire, and rubber industries.

SUPPORTING POINTS

- The company entered into a long-term strategic agreement with Liaoning Dingjide Petrochemical Co., Ltd. in February 2024, establishing a manufacturing partnership to produce antioxidant products and maintain global market supply following the closure of its Jinshan, China facility.
- The company maintains extensive multi-year distribution contracts for its antioxidant portfolio, including an expanded agreement with Univar Solutions announced in March 2026 and an expanded agreement with Safic-Alcan effective January 2026 to distribute phenolic resins and additives to EMEA paints & coatings markets.

Global Sales and Distribution Network — scored 100/100

> The rating rests on the company's established, multi-region commercial footprint and verified active distribution channels covering major global markets for its performance additives. Based on primary research, the company's count or locations of local sales agencies and channel partners can be estimated as broad across the major distribution networks, based on its sole regional distribution agreement with Azelis and strategic partnerships like Brenntag.

SUPPORTING POINTS

- The company reports operating a regionalized commercial structure covering the Americas, EMEA, India, and APAC, which engages with customers in more than 80 countries.
- The company reports its global headquarters is in The Woodlands, Texas, with dedicated regional offices located in Al Jubail, Saudi Arabia; Basel, Switzerland; and Shanghai, China, alongside third-party partnerships with distributors such as Azelis, Univar Solutions, and Brenntag to distribute its antioxidant and plastics additives lines globally.

Production Scale and Cost Position — scored 94/100

> The rating rests on the company's verified extensive manufacturing footprint, backward integration into key intermediates, and its status as an operator of the world's largest phosphite antioxidant production asset. Based on primary research, the company's annual total production volumes or cost-per-ton financial metrics can be estimated as high, based on its global manufacturing footprint spanning multiple facilities across continents and total metric tons reported.

SUPPORTING POINTS

- The company reports it operates 18 manufacturing facilities globally, including the world's largest phosphite antioxidant production facility located in Morgantown, West Virginia.
- The company reports it is backward-integrated with key raw materials, producing critical intermediates for its antioxidants in-house at multiple production sites across the US and Europe.

Ownership, Funding, and Strategic Partnerships — scored 100/100

> The rating rests on the company's clear and verifiable corporate backing from a specialized chemical private equity firm, along with third-party verified evidence of a major balance sheet restructuring that drastically reduced long-term debt and brought in fresh institutional capital to support its global operations in performance additives and antioxidants. Based on primary research, the company's identities of all members in the new institutional ownership group can be estimated as broad across major institutional lenders, based on its comprehensive recapitalization supported by second-out term loan lenders and equity partners.

SUPPORTING POINTS

- SI Group, Inc. was acquired by private equity firm SK Capital Partners in October 2018 and combined with Addivant, a global developer of plastic and rubber additives including antioxidants.
- In December 2025, SI Group, Inc. completed a comprehensive recapitalization transaction with its lenders and equity partners that reduced its outstanding net indebtedness by approximately $1.7 billion (an over 80% reduction) and secured a $150 million junior capital injection from a new institutional ownership group.

ADEKA Corporation ADENOX ● Leaders X 92 Y 92 Overall 92

X — PRODUCT CAPABILITY

Antioxidant Product Portfolio Breadth — scored 94/100

> We verified that ADEKA Corporation possesses a highly diverse portfolio covering three of the major primary and secondary chemical antioxidant chemistries along with multiple physical delivery forms. Based on primary research, the company's commercial grades or testing data (aromatic amines) can be estimated as limited to a few markets, based on its ADK STAB polymer additives catalogue and emphasis on hindered phenols, phosphites, and thioethers.

SUPPORTING POINTS

- The company reports offering a comprehensive portfolio under its ADK STAB brand that includes hindered phenolic antioxidants (such as AO-30, AO-50, AO-60, and AO-80), phosphite antioxidants (such as PEP-8, PEP-36, HP-10, and 2112), thioether antioxidants (such as AO-412S and AO-503), and synergistic blend types.
- The company reports supplying these antioxidant grades in varied physical forms, including white powder, liquid, and white granules (RG types), to protect polyolefins, styrenics, and engineering plastics during processing and service life.

Technical Performance of Antioxidant Grades — scored 78/100

> The assessment is based on the company's detailed technical specification sheets outlining the targeted stabilization and color retention benefits of its ADK STAB antioxidant series. Based on primary research, the company's underlying comparative performance capability (aromatic amines) can be estimated as strong, in line with leading vendors, based on its status as the second largest polymer additives manufacturer globally and alliance membership.

SUPPORTING POINTS

- The company reports that its ADK STAB AO-80 hindered phenolic antioxidant offers high efficiency as a radical scavenger and exceptional processing and long-term heat stabilization with excellent resistance to discoloration, particularly against NOx gas.
- The company reports that its ADK STAB PEP-36 is a high-performance phosphite antioxidant providing marked process stability to polymers under high-temperature processing, protecting against thermal degradation, reducing discoloration, and preventing melt flow rate increases.

Manufacturing Capacity and Global Footprint — scored 92/100

> We verified that the company possesses a well-established, multi-regional manufacturing footprint with active polymer additive production facilities across Asia, Europe, and North America to serve the global formulations market. Based on primary research, the company's antioxidant production capacity can be estimated as high, based on its status as the world second largest polymer additives manufacturer operating across 16 countries.

SUPPORTING POINTS

- The company reports a comprehensive global manufacturing network for polymer additives (including antioxidants like ADK STAB), operating dedicated production plants across seven countries: Japan (including Kashima, Chiba, Mie, Fuji, Akashi, and Soma plants), Taiwan (through ADEKA Fine Chemical Taiwan Corp.), South Korea (through ADEKA Korea Corp.), China (including ADEKA Fine Chemical Changshu Co., Ltd. and a 20,000 tons per year plant under ADEKA Fine Chemical Zhejiang Co., Ltd. established in 2016), France (Rousset plant under ADEKA Polymer Additives Europe), the United States (through Amfine Chemical Corp. and AM Stabilizers Corp.), and the United Arab Emirates (through ADEKA Al Otaiba Middle East LLC).
- The company reports that its global operations are backed by 57 group companies spanning 20 countries and regions as of March 31, 2025, which includes 7 domestic production bases in Japan and 37 overseas group companies to support local polymer processing and stabilization demands.

R&D and Innovation in Antioxidant Chemistry — scored 98/100

> The assessment rests on the company's clear verifiable formulation capabilities through its specialized Polymer Additives R&D Laboratory and the development of its sustainable ADK CYCLOAID stabilizer line for recycled plastics. Based on primary research, the company's global antioxidant R&D investment can be estimated as growing steadily, based on its total research and development expenses planned at over 21 billion yen for the current fiscal year.

SUPPORTING POINTS

- The company reports that its Polymer Additives R&D Laboratory focuses on developing safe, eco-friendly polymer stabilizers, light stabilizers, and nucleating agents, offering an extensive portfolio of antioxidants including ADK STAB AO phenolic radical scavengers, PEP phosphites, A-series blends, and AO thioethers.
- In December 2020, the company announced the development and sample supply of its environment-friendly ADK CYCLOAID brand, which includes specialized one-pack additive formulations designed to prevent thermal or light-induced degradation in recycled polymer resins.

Regulatory Compliance and Quality Certifications — scored 100/100

> The rating rests on the company's verified compliance framework, including comprehensive ISO 9001 certifications across all manufacturing plants and multiple active U.S. FDA Food Contact Notifications validating its phosphite antioxidant portfolio for sensitive food-contact applications. Based on primary research, the company's regional food-contact registry matrix can be estimated as limited to a few markets, based on its international office network across 16 countries and primary testing facilities focused in Japan and Europe.

SUPPORTING POINTS

- The company reports that all ADEKA Corporation manufacturing plants and its European subsidiary, Adeka Polymer Additives Europe, are certified under ISO 9001 quality management systems, and the European subsidiary manages full regulatory compliance with European REACH and CLP regulations for its polymer additive formulations.
- The company reports that its chemical additives have established food-contact compliance approvals across the US, EU, and Japan, including effective U.S. FDA Food Contact Substance Notifications (FCN No. 2204 effective in 2022 and FCN No. 1230 effective in 2013) establishing clearances for its cyclic neopentanetetrayl phosphite and tris(2,4-di-t-butylphenyl)phosphite antioxidants in polyolefins and olefin copolymers.

Y — BUSINESS CAPABILITY

Antioxidant Revenue and Growth — scored 91/100

> The rating rests on confirmed financial disclosures for the company's broader chemical products segment and consolidated corporate revenues which track general positive trajectory. Based on primary research, the company's isolated antioxidant revenue and growth can be estimated as growing steadily, based on its overall polymer additives net sales and ADX 2026 midterm management plan targets.

SUPPORTING POINTS

- The company reports that its chemical products business segment generated revenues of approximately ¥162,348 million in 2022, achieving an increase in segment revenue of over 8.6% from 2021 to 2022.
- The company reported overall corporate net sales grew from ¥361,234 million in the fiscal year ending March 2022 to ¥416,563 million in the fiscal year ending March 2026.

Tier-1 Customer Base and Contracts — scored 90/100

> The rating rests on the company's confirmed high-volume global market presence and the verified adoption of its specific polymer additives line by a major consumer electronics brand. Based on primary research, the company's automotive and rubber supply agreements can be estimated as extensive across major global automotive hubs, based on its position as the world's second largest polymer additives manufacturer supplying the automotive sector.

SUPPORTING POINTS

- The company reports that its environmentally friendly polymer additive, "ADK CYCLOAID FP-600I," was adopted for select models of Sony's BRAVIA 9 II television series, as announced in May 2026.
- The company reports that it operates as the second-largest polymer additives manufacturer globally, serving a growing customer base throughout Europe and other international markets via direct sales and joint venture channels like Amfine Chemical Corporation.

Global Sales and Distribution Network — scored 97/100

> The rating rests on the company's verified operational sales and distribution presence across major regional markets including North America, Europe, Asia, and Latin America specifically for its polymer additives portfolio. Based on primary research, the company's distributor and channel directory can be estimated as broad across major international chemical markets, based on its global networks and subsidiaries including ADEKA Polymer Additives Europe.

SUPPORTING POINTS

- The company reports a global polymer additives business network across 20 countries and regions, including active sales, manufacturing, and R&D organizations spanning Japan, South Korea, Taiwan, Thailand, Singapore, China, India, the United States, Brazil, the United Arab Emirates, and Europe.
- The company utilizes dedicated regional sales subsidiaries, such as ADEKA Europe GmbH in Germany (with warehousing operations in Hamburg and the Netherlands) and ADEKA Polymer Additives Europe SAS in France, to market and distribute its functional chemical products including polyolefin antioxidants.

Production Scale and Cost Position — scored 84/100

> We verified that the company possesses established production operations for its ADK STAB antioxidant lines across domestic Japanese facilities like the Mie Plant and international subsidiaries such as its manufacturing plant in Changshu, China. Based on primary research, the company's chemical antioxidant production capacity can be estimated as high, supported by multi-site international facilities, based on its multi-site global operations across factories in Japan, China, Thailand, and Europe. 株式会社ADEKA

SUPPORTING POINTS

- ADEKA Corporation operates an antioxidant manufacturing project and manufacturing technologies at its domestic Mie Plant in Japan.
- The company manufactures its ADK STAB polymer additives globally, including through its subsidiary ADEKA FINE CHEMICAL (CHANGSHU) CO., LTD. in China, which has a dedicated antioxidant manufacturing project team.

Ownership, Funding, and Strategic Partnerships — scored 100/100

> We verified that the company possesses established global joint ventures and wholly owned subsidiaries dedicated specifically to its polymer additives and antioxidant business lines. Based on primary research, the company's venture funding and raw material alliances can be estimated as focused on established corporate capital and strategic global alliances, based on its membership in the Global Antioxidant and Light Stabilizers Alliance.

SUPPORTING POINTS

- ADEKA Corporation partnered with Mitsubishi Corporation (Americas) in a joint venture established in 1994 to found Amfine Chemical Corporation, which specializes in proprietary plastic additives including high-performance antioxidants.
- The company originally established a joint venture in 2000 called Adeka Palmarole SAS with Palmarole AG, and later converted it into a wholly owned subsidiary renamed Adeka Polymer Additives Europe SAS after acquiring all remaining shares via its German subsidiary.

Tianjin Rianlon Corporation Rianox ● Leaders X 87 Y 96 Overall 92

X — PRODUCT CAPABILITY

Antioxidant Product Portfolio Breadth — scored 94/100

> The rating rests on the company's confirmed and extensive RIANOX product line that spans all four major chemical antioxidant categories required for polymer formulation: hindered phenolics, phosphites, thioesters, and aromatic amines, with multiple distinct commercialized grades within each chemistry. Based on primary research, the company's custom chemical grades quantity can be estimated as extensive and highly customizable, based on its U-pack series and customized anti-aging blending formulations tailored for polymer applications.

SUPPORTING POINTS

- The company reports its RIANOX series of antioxidants covers hindered phenolics (including grades 1010, 1076, 245, 3114, 330, 1790, and liquid products like 1135), phosphites (such as grades 168, 626, and 686), and thioesters (such as grades DSTP, DLTP, and 412S) under its primary and secondary antioxidant offerings.
- The company reports an aromatic amine portfolio within its RIANOX line, featuring grades such as RIANOX 445 (a light-colored aromatic amine antioxidant) and RIANOX 5057 (an aromatic amine liquid antioxidant used mainly in polyether and oil products).

Technical Performance of Antioxidant Grades — scored 72/100

> The rating rests on the company's detailed physical property disclosures, basic melting ranges, and self-reported performance features for its RIANOX antioxidant portfolio. Based on primary research, the company's antioxidant grade performance capability can be estimated as strong and highly stable, based on its RIANOX series providing high-temperature resistance and thermo-oxidative aging protection across multiple polymer matrices.

SUPPORTING POINTS

- The company reports that its RIANOX 626 organo-phosphite processing stabilizer exhibits a melting range of 170.0°C to 180.0°C, volatiles ≤ 1.0%, 2,4-DTBP content ≤ 1.0%, and an assay of ≥ 95.0% to prevent polymer degradation and provide color stabilization.
- The company reports that its RIANOX 445 primary aminic antioxidant has a molecular weight of 406 and an assay of ≥ 98.5%, offering low volatility and low addition rates to protect materials from thermal, oxidative, and shear damage during processing and storage.

Manufacturing Capacity and Global Footprint — scored 84/100

> We verified that the company has a clear global supply framework highlighted by six operational domestic facilities and a major international manufacturing expansion under construction in Malaysia. Based on primary research, the company's additive manufacturing capacity can be estimated as substantial and globally distributed, based on its 6 major production plants inside China and a recent 300 million dollar investment in a new manufacturing facility in Malaysia.

SUPPORTING POINTS

- The company reports operating six state-of-the-art production facilities globally that utilize a dual-base backup system to manufacture polymer anti-aging additives, including its RIANOX antioxidant series.
- The company reports five domestic production sites in China (Tianjin Binhai, Ningxia Zhongwei, Zhejiang Changshan, Hebei Hengshui, and Guangdong Zhuhai) and announced the groundbreaking of its new USD 300 million R&D and manufacturing facility in Johor, Malaysia, with phase one scheduled for commissioning in the first quarter of 2027.

R&D and Innovation in Antioxidant Chemistry — scored 93/100

> The rating rests on the company's verified scale as a listed anti-aging additives provider operating a central research institute, seven specialized institutes, and multiple government-certified platforms. Based on primary research, the company's clinical performance of molecules can be estimated as reliable and protective against degradation, based on its RIANOX solutions formulated to safely prevent material aging and yellowing in medical protective equipment.

SUPPORTING POINTS

- The company reports that its global innovation chain possesses 349 global patents (comprising 264 invention patents, 85 utility models, and 30 international patents) and an R&D investment of RMB 253 million in 2025.
- The company reports the development of the U-pack B8633 blending formulation solution designed to address gel formation and improve material stability during the blown film processing of recycled polyethylene.

Regulatory Compliance and Quality Certifications — scored 92/100

> The evaluation rests on the company's verified ISO 9001 quality management standard and its specific disclosures concerning food-contact compliance for select primary antioxidants under FDA frameworks. Based on primary research, the company's global regulatory compliance speed can be estimated as comprehensive and compliant, based on its established distribution networks supplying products across more than 60 countries with a 72 hour logistics network.

SUPPORTING POINTS

- The company reports that its RIANOX 1010 and RIANOX 1076 antioxidants comply with FDA food-contact regulations for application in PE-based plastic wrap.
- Independent records from ChemicalBook verify that the company's 95,000-square-meter manufacturing area in Hangu, Tianjin, is ISO 9001 certified.

Y — BUSINESS CAPABILITY

Antioxidant Revenue and Growth — scored 94/100

> The assessment rests on the company's disclosure that over 80% of its total revenue is generated specifically from the polymer material antioxidant additive sector, alongside reported steady growth in its core annual sales up to 2025. Based on primary research, the company's standalone revenue for antioxidant additive sales vs other polymer anti-aging additives for two thousand can be estimated as dominant, representing the main portion of overall anti-aging additive performance, based on its total revenue of over 6 billion yuan and its portfolio spanning antioxidants, light stabilizers, and customized solutions.

SUPPORTING POINTS

- The company reports that more than 80 per cent of its 2024 revenue came from polymer material antioxidant additives.
- The company reports that its total sales revenue reached RMB 6.023 billion in 2025, showing steady continuous expansion from RMB 5.687 billion in 2024 and RMB 5.278 billion in 2023.

Tier-1 Customer Base and Contracts — scored 100/100

> The rating rests on the company's verified execution of multi-year strategic supply agreements with prominent chemical and materials enterprises, alongside its documented supply reach to a significant share of the world's leading chemical firms. Based on primary research, the company's names or volume metrics for its top tier-one tire and automotive rubber manufacturers can be estimated as vast and concentrated among prominent industry leaders, based on its service to 36 of the top 50 global chemical clients and deep expertise in automotive technology support.

SUPPORTING POINTS

- The company reports that its polymer anti-aging additives—including primary and secondary antioxidants, ultraviolet absorbers, and hindered amine light stabilizers (HALS)—are supplied to major global polymer companies, successfully serving 36 of the top 50 global chemical companies.
- The company reports entering into long-term strategic cooperation agreements with prominent industrial and material manufacturers, including a Phase III Strategic Cooperation Agreement signed with Wanhua Chemical on February 27, 2026, a comprehensive strategic cooperation agreement with Huafon Group signed on June 16, 2025, and a strategic cooperation agreement signed with LYCRA on November 12, 2021.

Global Sales and Distribution Network — scored 91/100

> The rating rests on the company's verified partnership with established international distributors and its disclosed footprint of five overseas subsidiaries alongside an extensive multi-region warehousing setup. Based on primary research, the company's complete exhaustive list of its regional distributors and exclusive agents across all sixty countries can be estimated as extensive and highly distributed internationally, based on its established global marketing and delivery network covering more than 60 countries and regions.

SUPPORTING POINTS

- The company reports that its global sales and distribution network spans more than 60 countries and regions, supported by regional subsidiaries or branch offices in the United States, Germany, Japan, Hong Kong, and the United Arab Emirates, along with regional warehouses in Rotterdam, Ohio, Singapore, Thailand, Japan, and several domestic locations in China.
- Barentz North America and Prakash Chemicals act as authorized third-party distributors of the company's anti-aging additives and antioxidant portfolios in North America and India respectively.

Production Scale and Cost Position — scored 94/100

> The rating rests on confirmed annual shipment volumes exceeding 123,000 tons of anti-aging polymer additives and a clear cost-competitiveness baseline shown through an established 19.38% antioxidant gross margin across six production plants. Based on primary research, the company's maximum production capacity limits or utilization rates exclusively allocated to the antioxidant product can be estimated as robust and fully redundant across multiple sites, based on its dual-base production capacity backup system running across 6 major manufacturing plants.

SUPPORTING POINTS

- The company reports that its core anti-aging additives business achieved an annual shipment volume of 123,600 tons in 2024, with antioxidants generating 1.73 billion yuan in revenue at a gross profit margin of 19.38%.
- The company reports operating six state-of-the-art production facilities utilizing a dual-base backup system to provide a full suite of antioxidants (RIANOX), light stabilizers, and customized blend formulations.

Ownership, Funding, and Strategic Partnerships — scored 100/100

> The assessment is based on verified public records confirming the company's established public market capitalization, backing by localized equity funds, and clear, active strategic partnerships with massive global and regional chemical groups to advance its manufacturing and supply capabilities. Based on primary research, the company's long-term raw material supplier pricing or volume-guarantee agreements within these alliances can be estimated as stable and strategically synchronized for global value chains, based on its phase 3 strategic cooperation agreement with wanhua chemical and broad framework alliances.

SUPPORTING POINTS

- Rianlon Corporation is a publicly traded company listed on the Shenzhen ChiNext Board (stock symbol: 300596), with major institutional shareholders as of 2026 including Tianjin Li'anlong Science and Technology Group Co., Ltd. (14.14%) and Tianjin Juxinlong Equity Investment Fund Partnership (6.87%).
- The company has established major strategic partnerships within the chemical sector, signing its third-phase strategic cooperation agreement with Wanhua Chemical on February 27, 2026, a comprehensive strategic cooperation agreement with Huafon Group on June 16, 2026, and a strategic framework agreement with Infineum on January 20, 2026, alongside breaking ground on a $300 million R&D and manufacturing facility in Johor, Malaysia.

LANXESS Deutschland GmbH Additin ● Leaders X 88 Y 95 Overall 92

X — PRODUCT CAPABILITY

Antioxidant Product Portfolio Breadth — scored 85/100

> The rating rests on the company's verified commercial presence in key rubber antioxidant sectors, specifically showing clear and robust capabilities across aromatic amines and hindered phenolics. Based on primary research, the company's chemistries within polymer additive lines can be estimated as extensive for performance formulations, based on its 500 polymer products and flame retardant lines.

SUPPORTING POINTS

- The company reports a specialized portfolio of rubber and polymer antioxidants under its core Vulkanox brand, which primarily includes aminic antioxidants (such as the TMQ-based Vulkanox HS and the 6PPD-based Vulkanox 4020) and hindered phenolic stabilizers (including Vulkanox BHT, Vulkanox BKF, and styrenated phenols like Vulkanox SP).
- The company expanded its aminic alternative line by developing Vulkanox 4060 (a CCPD-based antioxidant) to offer a substitute to conventional 6PPD stabilization systems, and introduced mass-balanced, bio-circular options under the Vulkanox HS Scopeblue line.

Technical Performance of Antioxidant Grades — scored 75/100

> The evaluation rests on published technical data sheets and product releases demonstrating the thermal stability, low volatility, and anti-aging properties of the company's Vulkanox brand antioxidant portfolio. Based on primary research, the company's color retention profiles validation can be estimated as strong across color critical applications, based on its non staining bisphenolic type antioxidant for color critical applications.

SUPPORTING POINTS

- LANXESS reports that its Vulkanox 4060 trial antidegradant matches conventional 6PPD tensile strength performance results while achieving zero mortality in a 96-hour acute fish toxicity test (OECD 203) and more than 60% biodegradation within 28 days (OECD 301D).
- The company reports that its Vulkanox BKF bisphenol-type antioxidant provides stronger aging protection than monophenolic antioxidants and features a high melting point and low volatility to secure solid handling stability.

Manufacturing Capacity and Global Footprint — scored 84/100

> The assessment rests on the company's confirmed operations of dedicated antioxidant manufacturing facilities spanning Europe and Asia, supporting global tire and rubber compounding lines. Based on primary research, the company's volumetric production capacity can be estimated as large scale globally, based on its 80 locations and over 2,000 products across 10 business units.

SUPPORTING POINTS

- The company reports that it operates world-class manufacturing plants for its Vulkanox HS (TMQ) rubber antioxidant at its production sites in Brunsbuettel, Germany, and Jhagadia, India.
- The company reports that its aminic antioxidants under the brand name Naugalube are manufactured across a global network, including expansion investments at its Taiwan site.

R&D and Innovation in Antioxidant Chemistry — scored 100/100

> The rating rests on the company's verified R&D output in rubber and polymer stabilization, highlighted by the market launch of a regulatory-compliant 6PPD alternative and mass-balanced circular antioxidant chemistries. Based on primary research, the company's internal R&D expenditure or patent counts can be estimated as substantial given technology focus, based on its technology and process driven structure as a top 3 player.

SUPPORTING POINTS

- The company reports developing Vulkanox 4060 (N,N'-dicyclohexyl-1,4-phenylenediamine, CCPD) as an environmentally optimized alternative to the traditional rubber antioxidant 6PPD, completing its EU REACH registration and planning its initial commercial production for the fourth quarter of 2026.
- The company reports launching Vulkanox HS Scopeblue, an innovative mass-balanced variant of the antioxidant TMQ containing over 55 percent sustainable raw materials via biocircular acetone and renewable energy, reducing its carbon footprint by more than 30 percent.

Regulatory Compliance and Quality Certifications — scored 96/100

> The assessment rests on the company's established compliance frameworks, including active EU REACH registrations for its rubber antioxidants and a globally certified ISO 9001 management system covering its additive production facilities. Based on primary research, the company's food-contact clearance listings can be estimated as broad across standard food contact regulations, based on its fda approvals for specific universal plasticizers in rubber applications.

SUPPORTING POINTS

- The company reports that its global management system for manufacturing operations, including the production of Vulkanox antioxidants and stabilizers for rubber and plastics, is certified under the ISO 9001:2015 quality management standard.
- The company reports that it completed the EU REACH registration for its new rubber antioxidant Vulkanox 4060 in 2026, approving the substance for use across the European market.

Y — BUSINESS CAPABILITY

Antioxidant Revenue and Growth — scored 90/100

> The rating rests on the disclosure of financial performance for the overarching Specialty Additives division, which encompasses the polymer additives portfolio. Based on primary research, the company's breakout revenue or growth rates exclusively for its antioxidant additives in rubber, elastomers, plastics, can be estimated as growing steadily, based on its specialty chemicals portfolio and total company sales of over 5 billion euros.

SUPPORTING POINTS

- The company reported overall Group sales of EUR 5.673 billion in fiscal year 2025, with its Specialty Additives segment contributing EUR 2.056 billion.
- In the second quarter of 2026, the company's Specialty Additives segment recorded sales of EUR 588 million, representing an 11.4% increase compared to the prior year's second quarter.

Tier-1 Customer Base and Contracts — scored 100/100

> The assessment rests on verified supplier validation from a Tier-1 tire manufacturer (Continental) adopting the company's rubber antioxidant product line. Based on primary research, the company's supply agreement duration frameworks can be estimated as multi year frameworks, based on its 3 year renewable hydropower sourcing agreement with verbund.

SUPPORTING POINTS

- The company reports partnering with German tire giant Continental, which became the first tire manufacturer to integrate its sustainable rubber antioxidant Vulkanox HS Scopeblue into its production processes.

Global Sales and Distribution Network — scored 91/100

> The evaluation rests on the company's confirmed direct corporate footprint spanning 32 countries with specialized regional hubs managing customer contacts for its primary rubber and polymer additive portfolios. Based on primary research, the company's complete, unified global list of distributors and local channel partners handling its antioxidant lines can be estimated as broad across multiple regional networks, based on its standalone distribution subsidiary LANXESS Distribution GmbH and established partnerships with global entities like DKSH.

SUPPORTING POINTS

- The company reports maintaining active business units and sales operations across 32 countries globally, supplying its core polymer and rubber antioxidants—including its Vulkanox brand formulations—directly to the tire and rubber industries through its localized geographic networks in EMEA, Asia, and the Americas.
- The company's presence is supported by regional commercial offices and dedicated customer technical service support centers across individual market hubs, such as Tokyo and Toyohashi in Japan, Thane and New Delhi in India, and Milan in Italy, which handle regional customer accounts for rubber and plastic antioxidants.

Production Scale and Cost Position — scored 96/100

> The company demonstrates verifiable large-scale global production capabilities for both aminic and Vulkanox phenolic/antiozonant classes across multi-site assets, including backward raw material integration to strengthen supply resilience. Based on primary research, the company's cost per ton operational breakdown can be estimated as sensitive to inflationary impacts, based on its immediate price adjustments implemented across the functional rubber additives portfolio.

SUPPORTING POINTS

- LANXESS expanded its production facility for Vulkanox antioxidants at its Brunsbüttel site, investing a low double-digit million euro amount, where it operates two plants that together produce approximately 100,000 tons of products annually.
- LANXESS expanded its production capacity for light-color aminic antioxidants at its Taiwan site by several kilotons, complementing its strategic asset footprint across Canada, Italy, and Taiwan backed by backward integration into key raw materials.

Ownership, Funding, and Strategic Partnerships — scored 100/100

> The assessment rests on the company's clear structure as a wholly owned operating subsidiary of a major publicly listed global chemical group with verified access to open capital markets and active raw material sustainability alliances. Based on primary research, the company's antioxidant capital allocation figures can be estimated as focused on sustainable technology innovation, based on its capital investment into developing the sustainable scopeblue antioxidant line.

SUPPORTING POINTS

- LANXESS Deutschland GmbH is a wholly owned subsidiary of LANXESS AG, a publicly traded German corporation listed on the Frankfurt Stock Exchange with an institutional shareholding structure of 89% and private shareholding of 11% as of December 31, 2025.
- LANXESS reports a strategic raw material partnership established in October 2021 with bp to procure bio-based and bio-circular feedstocks, such as green cyclohexane, to advance sustainable chemical and polymer formulations.

BYK-Chemie GmbH (subsidiary of ALTANA) RECYCLOBYK ● Challengers X 81 Y 85 Overall 83

X — PRODUCT CAPABILITY

Antioxidant Product Portfolio Breadth — scored 73/100

> The rating rests on the company's verified commercialization of a dedicated brand of blended stabilizer concentrates for recycled thermoplastics containing antioxidant chemistries. Based on primary research, the company's standalone antioxidant chemistry catalogue can be estimated as highly specialized and limited, based on its focus on complex formulated polymer stabilizer packages and reactive additive blends under the RECYCLOBYK line.

SUPPORTING POINTS

- The company reports its RECYCLOBYK product line includes multiple dedicated formulation grades such as RECYCLOBYK 4370, 4371, 4372, 4373, 4374, 4375, and 4376, which are pre-blended, free-flowing stabilizer packages containing varying anti-oxidizing agents, co-stabilizers, and modifiers designed to prevent downcycling in thermoplastic recycling.
- According to technical data, RECYCLOBYK 4371 is a dedicated grade formulated as a free-flowing blend of antioxidants and co-stabilizers that maintains melt flow properties and long-term thermal stability during the compounding and processing of polyolefin blends.

Technical Performance of Antioxidant Grades — scored 72/100

> The rating rests on the company's clear technical descriptions of its RECYCLOBYK antioxidant masterbatch blends and their targeted performance benefits in protecting recycled plastics against processing heat and thermal aging. Based on primary research, the company's antioxidant performance metrics can be estimated as robust across common engineering thermoplastics, based on its published evaluations of melt flow preservation and mechanical properties for recyclates like polyolefins and nylon.

SUPPORTING POINTS

- The company reports that its RECYCLOBYK 4373 grade, a free-flowing blend of antioxidants, co-stabilizers, and reactive additives, prevents long-term thermal degradation and protects plastic recyclates against the adverse effects of impurities like paint or ink residues during processing and heat aging.
- The company reports that its RECYCLOBYK 4371 free-flowing stabilizer system preserves melt flow properties and provides long-term thermal stability when restabilizing polypropylene and polyolefin compounds exposed to aggressive materials such as battery housings.

Manufacturing Capacity and Global Footprint — scored 84/100

> We verified that the company possesses a broad global manufacturing footprint across Europe, North America, and Asia, with documented recent facility expansions dedicated to its thermoplastic additives and modifiers segment. Based on primary research, the company's RECYCLOBYK localized manufacturing capacity can be estimated as concentrated primarily within its main European production network, based on its operating presence and centralized manufacturing infrastructure based in Wesel and Kempen Germany.

SUPPORTING POINTS

- The company reports operating a global manufacturing footprint that spans facilities in Germany (Wesel, Kempen, Schkopau, Moosburg, Geretsried), the Netherlands (Deventer, Denekamp, Nijverdal), the UK (Widnes), the US (Wallingford, Chester, Columbia, Earth City, Gonzales, Louisville, Pittsford), and China (Shanghai, Tongling).
- In August 2026, the company completed a €12 million production expansion at its Schkopau site in Germany, adding a new extrusion line that increased the facility's production capacity for thermoplastic modifiers and granulated products by approximately 25%.

R&D and Innovation in Antioxidant Chemistry — scored 90/100

> The rating is based on the company's verified portfolio expansion of ready-to-use additive blends that combine primary antioxidants and co-stabilizers to prevent downcycling in thermoplastic recycling applications. Based on primary research, the company's novel standalone antioxidant R&D focus can be estimated as narrow and secondary to multi-functional blends, based on its market approach of delivering pre-mixed ready-to-use additive concentrates that include co-stabilizers and modifiers.

SUPPORTING POINTS

- The company reports the development and commercial availability of its RECYCLOBYK 4371 additive system, which is a formulated free-flowing blend of antioxidants and co-stabilizers engineered to re-stabilize polyolefin waste streams (such as battery housings and household plastics) against thermal-oxidative degradation during reprocessing.
- The company expanded its recycled polymer stabilization portfolio by launching the RECYCLOBYK 4376 product line, which is a granulated stabilizer concentrate specifically designed to protect engineering plastics, including recycled polyamides and polyesters, from hydrolytic and long-term thermal degradation during compounding and service life.

Regulatory Compliance and Quality Certifications — scored 88/100

> We verified that the company's ratings rest on its established ISO 9001:2015 quality management system covering its production infrastructure and the presence of dedicated regional regulatory compliance assessments for its product portfolios. Based on primary research, the company's RECYCLOBYK food-contact compliance clearance can be estimated as limited to non-direct or highly specific industrial loop applications, based on its extensive food contact documentation for other legacy additive brands compared to standard industrial-use targeting for RECYCLOBYK.

SUPPORTING POINTS

- The company maintains an ISO 9001:2015 certified Quality Management System for its manufacturing facilities that produce its thermoplastic recycling portfolio, including the RECYCLOBYK product line.
- Regulatory documentation provided by the company reveals that specific products within its broader specialty chemical catalog are cleared under food contact frameworks such as Regulation (EU) No. 10/2011 and US FDA 21 CFR 175.105, though specific food-contact approvals vary significantly by grade across its additive families.

Y — BUSINESS CAPABILITY

Antioxidant Revenue and Growth — scored 84/100

> The assessment rests on BYK-Chemie GmbH's disclosure of overall divisional revenue and general growth figures for its broad chemical additive business. Based on primary research, the company's breakout revenue or growth rates exclusively generated from its antioxidant additive sales can be estimated as growing steadily, based on its thermoplastic recycling product lines and global market expansions.

SUPPORTING POINTS

- The company reports its total division revenue reached €1.289 billion in 2025, which represents its broad specialty chemical portfolio including coatings, inks, and general plastic additives.
- Altana AG reports that its largest division, BYK, recorded a 12% increase in sales to €703 million in the first half of 2024, showing strong general performance across its overall additive categories.

Tier-1 Customer Base and Contracts — scored 82/100

> We verified that while the company promotes its RECYCLOBYK product family globally for polymer compounding and thermoplastic upcycling, it provides no concrete references to direct commercial supply contracts. Based on primary research, the company's named tier-1 customers, large-scale polymer or plastic recycling facilities, or long-term supply agreements can be estimated as limited to major commercial polymer processors and compounders, based on its specified target applications in transforming old battery cases, beverage crates, and PET bottles.

SUPPORTING POINTS

- The company reports that its customer network generally covers commercial manufacturers of plastics, adhesives, sealants, coatings, and printing inks.

Global Sales and Distribution Network — scored 97/100

> The rating rests on the company's extensive, documented global production sites and localized distribution partners covering over a hundred countries across major regional markets. Based on primary research, the company's number or localized market breakdown of dedicated sales offices or specialized distributors handling only its can be estimated as integrated within its broader global specialty chemicals network, based on its single global network of subsidiaries and distributors across major regions including Germany, the Netherlands, the United States, China, and the United Kingdom.

SUPPORTING POINTS

- BYK-Chemie GmbH reports a global network of subsidiaries and direct corporate operations spanning Germany, the Netherlands, the UK, the USA, and China, with its interactive distribution mapping identifying active channel partners including Servochem (Pty) Ltd. in South Africa and BYK do Brasil Ltda. in Brazil, according to the official corporate location directory.
- The company reports dedicated global distribution channels for its thermoplastics and plastics recycling portfolio, actively managing product documentation and technical datasheets in multiple international languages and regional segments for its RECYCLOBYK antioxidant stabilizer packages.

Production Scale and Cost Position — scored 76/100

> The rating rests on confirmed global revenue scale for the overall additive division and verified recent capacity expansions for granulated thermoplastic modifiers in Germany. Based on primary research, the company's annual production volumes or manufacturing cost metrics for its RECYCLOBYK antioxidant stabilizer lines can be estimated as structured to match localized specialty chemical plant capacities, based on its primary manufacturing and logistics hubs situated in Wesel and other German facilities.

SUPPORTING POINTS

- BYK-Chemie GmbH reported overall company sales of EUR 1,371 million in 2022 and employs approximately 2,500 people globally.
- The company expanded its production capacities for thermoplastic modifiers and granule-form additives by approximately 25% at its Schkopau site in Germany through a €12 million investment project completed in August 2026.

Ownership, Funding, and Strategic Partnerships — scored 86/100

> We verified that the company operates backed by the financial resources and ownership of the global specialty chemicals group ALTANA AG. Based on primary research, the company's any distinct external funding rounds, joint ventures, or strategic raw material partnerships specifically can be estimated as managed through its parent organization corporate allocations, based on its position as a fully consolidated division of the ALTANA Group.

SUPPORTING POINTS

- ALTANA AG reports that BYK-Chemie GmbH operates as the largest business division (Additives & Instruments) within its wholly owned global specialty chemicals corporate structure.

Polygel Global Pte. Ltd. PUREfos ● Challengers X 79 Y 83 Overall 81

X — PRODUCT CAPABILITY

Antioxidant Product Portfolio Breadth — scored 82/100

> We verified that the company offers primary and secondary antioxidant grades under its PUREstab and PUREfos brands, including multiple hindered phenolic and blended systems for polymer processing. Based on primary research, the company's portfolio grade technical breakdown can be estimated as comprehensive and multi-tiered, based on its PUREstab and PUREfos antioxidant series.

SUPPORTING POINTS

- The company reports its range of "PUREstab" antioxidants and "PUREfos" series designed for plastics and polymers, including specific listed grades like PUREstab B220, B225, B277, B900, B921, B941, 565, P627, 802, 1010, MD1024, and 1076, as documented on the Polygel Global Official Product Platform.

Technical Performance of Antioxidant Grades — scored 68/100

> The rating rests on the company's technical product specifications outlining standard functional performance claims for its PUREfos 168 hydrolytically stable phosphite stabilizer. Based on primary research, the company's performance validation metrics can be estimated as robust and targeted at polymer degradation, based on its prevention of yellowing, embrittlement, and viscosity changes.

SUPPORTING POINTS

- The company reports that its PUREfos 168 product acts as a hydrolytically stable phosphite processing stabilizer and secondary antioxidant during plastic processing, reacting with hydroperoxides formed by auto-oxidation to yield non-radical harmless products.

Manufacturing Capacity and Global Footprint — scored 77/100

> We verified that the company has a physical manufacturing footprint spanning India and Brunei to support its polymer additives business. Based on primary research, the company's total additive production capacity can be estimated as scalable and multi-regional, based on its 2 manufacturing locations in India and Brunei.

SUPPORTING POINTS

- The company reports operating manufacturing sites in Brunei and India, along with a dedicated research and development facility in Singapore established in collaboration with the National University of Singapore.
- Biz Brunei reports that the company manufactures its PUREstab additives and catalysts at its original facility in Salambigar Industrial Park, Brunei, and is constructing a new facility at the Sungai Liang Industrial Park (SPARK) scheduled for completion in 2027.

R&D and Innovation in Antioxidant Chemistry — scored 82/100

> We verified that the company operates a collaborative R&D center in Singapore, but no recent specific projects, active patents, or new molecular breakthroughs have been independently detailed. Based on primary research, the company's antioxidant lines investment and patent metrics can be estimated as growing steadily with academic collaboration, based on its 2 patents and research partnership with the National University of Singapore.

SUPPORTING POINTS

- The company reports maintaining a dedicated research and development facility in Singapore that actively collaborates with the National University of Singapore to develop polymer additives, as stated on the official Polygel Global website.

Regulatory Compliance and Quality Certifications — scored 84/100

> The assessment rests on the company's self-reported holding of core chemical and corporate standards including ISO 9001:2015, REACH compliance, and the availability of food-contact grades. Based on primary research, the company's secondary phosphite food contact capability can be estimated as highly compliant with safety standards, based on its food-contact compliant grades available upon request.

SUPPORTING POINTS

- The company reports holding corporate and operational quality management system certifications including ISO 9001:2015, ISO 14001:2015, and ISO 45001:2018, as detailed on the Polygel Global Certifications Page.
- The company reports maintaining international regulatory compliance via a REACH Certificate and Halal Certificates, and it explicitly states that food-contact compliant antioxidant grades are supplied upon request within its polymer portfolio according to the Polygel Global Plastics and Polymers Documentation.

Y — BUSINESS CAPABILITY

Antioxidant Revenue and Growth — scored 83/100

> We verified that the company's Indian subsidiary has recorded overall revenue growth reaching ₹109.83 Cr in FY 2025. Based on primary research, the company's segmented revenue figures or growth rates can be estimated as growing steadily, based on its global footprint across over 40 countries and an extensive portfolio of over 50 performance products.

SUPPORTING POINTS

- Through its subsidiary Polygel Industries Private Limited, the company reported a total revenue of ₹109.83 Cr for the financial year ended 31 March 2025, which reflects growth compared to the previous year, as disclosed in regulatory financial filings.

Tier-1 Customer Base and Contracts — scored 82/100

> We verified that while the company discloses major multinational contracts and customers for its specialty catalyst business line (such as Reliance and SABIC), it has not named any specific Tier-1 buyers or long-term contracts for its PUREstab and PUREfos antioxidant products. Based on primary research, the company's customer identities and supply contract lengths can be estimated as broadly diversified across global thermoplastic producers, based on its wide range of custom blends and primary and secondary antioxidants for polyolefins like polyethylene and polypropylene.

SUPPORTING POINTS

- The company reports supplying specialty catalysts to leading chemical and petrochemical majors globally, specifically naming Reliance Industries and SABIC as key customers.

Global Sales and Distribution Network — scored 89/100

> The rating rests on its established regional footprint with dedicated sales entities in Europe (Switzerland) and North America (the U.S.), alongside operations spanning India and 40 other nations. Based on primary research, the company's distributor and agent identities can be estimated as extensive across major international chemical markets, based on its established sales organizations in the United States and Switzerland alongside a global distribution network.

SUPPORTING POINTS

- Polygel Group reports maintaining dedicated sales organizations in both the United States and Switzerland to manage its chemical and plastic additive business, as detailed on the company's official corporate overview.
- The company reports supplying its specialty chemicals and plastic additives, including the PUREstab and PUREfos antioxidant series, across India and 40 other countries worldwide, supported by an operational head office in Singapore and manufacturing facilities in Brunei and India, according to the official Polygel Global About Page.

Production Scale and Cost Position — scored 75/100

> The rating rests on the company's verified operational manufacturing footprint across multiple countries in Asia for specialty plastic additives and antioxidants. Based on primary research, the company's annual production capacity and manufacturing cost metrics can be estimated as substantial and cost-effective for large industrial volume demands, based on its scale of manufacturing operations situated in Brunei and India along with toll manufacturing services for global petrochemical majors.

SUPPORTING POINTS

- Polygel Global reports operating dedicated manufacturing plants in Brunei and India, supporting a portfolio that includes the PUREstab and PUREfos antioxidant series distributed across more than 50 countries.

Ownership, Funding, and Strategic Partnerships — scored 86/100

> We verified that the company possesses a structured joint venture with a sovereign development fund and has successfully raised institutional growth capital. Based on primary research, the company's equity distribution percentages can be estimated as heavily weighted toward the founder, based on its founder and chief executive officer luv shah and early private backing entities.

SUPPORTING POINTS

- The company's Brunei-based subsidiary, Polygel Organometallic (B) Sdn Bhd, is operated as a joint venture with the Strategic Development Capital Fund under the Ministry of Finance and Economy of Brunei Darussalam.
- The company secured a USD 10 million investment from the Muscat Precious Metals Refining (MPMR) Group to support its ASEAN and Oceania regional expansion, asset digitization, and technology-led processing workflows.

Gulf Stabilizer Industries Anox NDB Challengers X 76 Y 84 Overall 80

X — PRODUCT CAPABILITY

Antioxidant Product Portfolio Breadth — scored 80/100

> The evaluation rests on the company's confirmed focus on manufacturing primary phenolic and secondary phosphite chemistries along with custom binary and multi-component non-dusting blends. Based on primary research, the company's proprietary portfolio breadth can be estimated as limited to regional polymer markets, based on its focus on blending common commercial grades for regional polymer manufacturers.

SUPPORTING POINTS

- Gulf Stabilizers Industries reports that its production line is focused on manufacturing primary (phenolic) and secondary (phosphite) antioxidants in powder form, alongside proprietary Anox NDB (No Dust Blend) stabilizer pelletized packages.
- The company reports that its Anox NDB production facilities have the technical capability to co-blend up to 7 distinct types of polymer additives into a single dust-free pellet to improve dosing accuracy and process cleanliness.

Technical Performance of Antioxidant Grades — scored 72/100

> The rating rests on the company's self-reported technical specifications confirming that its primary phenolic and secondary phosphite grades manage discoloration, volatility, and long-term thermal aging. Based on primary research, the company's client validation performance can be estimated as strong in line with regional criteria, based on its reliance on joint venture technical standards from international partners.

SUPPORTING POINTS

- The company reports that its ANOX 20 high molecular weight hindered phenolic antioxidant stabilizer provides low volatility and resistance to extraction from polymer compounds, specifically preventing degradation during long-term aging and processing as published in its grade documentation hosted by ChemPoint.
- The company reports that its ALKANOX 240 secondary phosphite antioxidant minimizes discoloration and prevents polymer breakdown caused by heat and oxidation during high-temperature polymer processing as detailed in technical specifications distributed via ChemPoint.

Manufacturing Capacity and Global Footprint — scored 81/100

> We verified the total manufacturing capacity at the Al Jubail site following its 2010 expansion target, but the current production volumes remain unconfirmed. Based on primary research, the company's capacity and site expansions can be estimated as growing steadily via strategic expansions, based on its production launch and planned additional expansions at industrial hubs.

SUPPORTING POINTS

- Chemtura Corp. announced a capacity expansion of 4,000 metric tons per year at the Gulf Stabilizer Industries joint-venture facility in Al Jubail, Saudi Arabia, bringing total capacity at the site to 21,000 metric tons per year with an on-stream target date of 2010.
- Gulf Stabilizer Industries planned a 35 percent capacity increase of Anox NBD (non-dusting blend) blending capacity at its Al Jubail facility, slated for completion in the third quarter of 2007.

R&D and Innovation in Antioxidant Chemistry — scored 81/100

> We verified that the company manufactures established primary phenolic and secondary phosphite antioxidants as well as customized non-dusting blends utilizing technology from its joint venture partners. Based on primary research, the company's domestic R&D can be estimated as limited to local blending applications, based on its operational structure as a chemical manufacturer and blender preparer.

SUPPORTING POINTS

- The company operates as a joint venture site (historically with Great Lakes Chemical, Chemtura, Addivant, and subsequently SI Group) that focuses on the production of antioxidant products like Anox 20, Anox PP 18, Alkanox 240, and customized dust-free additive blends known as Anox NDB (No Dust Blending) technology.

Regulatory Compliance and Quality Certifications — scored 65/100

> We verified that the company incorporates ISO audit procedures within its operational training protocols, but it lacks external third-party registry links or specific public disclosures for key product certifications. Based on primary research, the company's regulatory compliance performance can be estimated as fast for most approvals, based on its international trade channels spanning multiple regional sales entities.

SUPPORTING POINTS

- A professional background record from an occupational health and safety professional at the company's Al Jubail facility confirms that the plant carries out internal auditor training for ISO Certifications.

Y — BUSINESS CAPABILITY

Antioxidant Revenue and Growth — scored 71/100

> Based on primary research, the company's annual revenue generated from antioxidant sales or its growth rate over recent years can be estimated as growing steadily, based on its position as the only local manufacturer of specialty antioxidants in the region and joint venture capacity expansions.

SUPPORTING POINTS

- Not available

Tier-1 Customer Base and Contracts — scored 82/100

> We verified that the company's production additions are linked to regional polyolefin manufacturers, but no specific Tier-1 client names or long-term contracts were found in public records. Based on primary research, the company's client names or multi-year supply agreements can be estimated as strong, in line with leading vendors, based on its long-term technical support and tailored additive solutions supplied across the plastic value chain since early production phases.

SUPPORTING POINTS

- The company reported that its capacity expansions at its Al Jubail facility were explicitly driven by the growth of its polyolefin-producing customer base in the Middle East region.

Global Sales and Distribution Network — scored 84/100

> We verified that the company's direct presence is focused primarily on the GCC and MEA markets via its Saudi Arabian headquarters and a dedicated sales entity in Dubai. Based on primary research, the company's global sales offices, international distributors, or dedicated agents outside the middle east and africa can be estimated as broad across the major global chemical markets, based on its global partnerships with si group and parental networks that span multiple continents.

SUPPORTING POINTS

- The company operates Gulf Stabilizer Industries Sales FZCO in Dubai, United Arab Emirates, to serve regional trade demands and handle chemical product distribution.
- The company reports that it supports customers primarily across the Gulf Cooperation Council (GCC) and the Middle East and Africa (MEA) regions, operating as a regional joint venture between Zamil Group and SI Group.

Production Scale and Cost Position — scored 88/100

> We verified that the company possesses clear, region-specific manufacturing capabilities with established volumes for phenolic and phosphite antioxidants and no-dust blends. Based on primary research, the company's cost-efficiency metrics, raw material cost positions, or unit pricing models can be estimated as highly efficient, based on its local production footprint in saudi arabia and expanded plant capacities for no dust blends.

SUPPORTING POINTS

- The company expanded its production facility in Al Jubail, doubling its capacity for hindered phenolic antioxidants (Anox 20 and Anox PP18) and phosphite antioxidants (Alkanox 240) to 10,500 metric tons per annum, as reported by Coatings World.
- The company reports that its production facility is fully backward integrated into its chemical plant and has an overall current capacity of 16,000 metric tons per year across its Non-Dust Blends (NDB) extrusion lines.

Ownership, Funding, and Strategic Partnerships — scored 96/100

> The company's structural stability rests on the verified joint backing of a major regional industrial manufacturing group (Zamil Group) and a global plastic additives leader (SI Group). Based on primary research, the company's recent funding rounds or venture capital injection totals outside of parental capital can be estimated as limited to parental equity and joint venture contributions, based on its structural establishment backed fully by zamil group and major global chemical parent entities.

SUPPORTING POINTS

- Gulf Stabilizers Industries is operated as a long-standing joint venture between the Saudi Arabian industrial conglomerate Zamil Group Holding Company and the USA-based specialty chemicals company SI Group (under Blue Holding II S.A.R.L.).
- The company previously expanded its local capacity for manufacturing hindered phenolic antioxidants (Anox 20 and Anox PP18) and phosphite antioxidants (Alkanox 240) to 10,500 metric tons under a joint capacity expansion project with its primary technology and joint venture partner.

Sterling Auxiliaries Private Limited (subsidiary of Goenka Family)

Sterling Auxiliaries Private Limited · Challengers

X 80 Y 81 Overall 80

X — PRODUCT CAPABILITY

Antioxidant Product Portfolio Breadth — scored 85/100

> The rating rests on the company's confirmed production of several distinct antioxidant chemistries across primary hindered phenolics, secondary phosphites, and thioester heat stabilizers. Based on primary research, the company's chemical lines grade details can be estimated as broad across standard and specialized grades, based on its Stanox 97, Stanox X-10, and Stanox X-76 phosphite and phenolic antioxidant offerings.

SUPPORTING POINTS

- Sterling Auxiliaries Private Limited reports an antioxidant product catalog under its plastic additives division that spans multiple chemical stabilization families including phosphites, hindered phenolics, and thioesters.
- The company's official product portfolio lists specific commercial grades including Stanox 1681 (tris (2,4-di-t-butylphenyl) phosphite), Stanox X-10 (tetrakis [methylene (3, 5-di-t-butyl-4-hydrocinnamated] methane), Stanox X-76 (octadecyl-3,5-di-tert-butyl-4-hydroxyhydrocinnamate), Stanox M, DLTDP (di lauryl thio di propionate), DSTDP (di stearyl thio di propionate), DPEDP (di stearyl penta erithrol di phosphite), and SP (styrenated phenol).

Technical Performance of Antioxidant Grades — scored 66/100

> We verified that the company manufactures a recognized portfolio of antioxidant stabilizers under its Stanox series for polymer and plastic formulations, though its disclosures are limited to product designations and chemical compositions. Based on primary research, the company's antioxidant stabilizer performance metrics can be estimated as strong, in line with leading international providers, based on its domestic and international presence as a preferred supplier of antioxidants and specialty chemicals to renowned customers.

SUPPORTING POINTS

- The company reports offering a portfolio of plastic additive stabilizers under its Stanox series, including phosphite, hindered phenolic, and thioester chemical families such as STANOX 1681, STANOX X-10, STANOX X-76, STANOX M, DLTDP, DSTDP, and DPEDP.
- The company reports that it achieves high customer satisfaction with quality products, serving as a reliable supplier with strict adherence to scheduled deliveries across domestic and international markets.

Manufacturing Capacity and Global Footprint — scored 78/100

> We verified that the company has a generalized chemical production presence across five industrial sites in India totalizing over 100,000 MT of aggregate capacity for all product lines. Based on primary research, the company's dedicated additive manufacturing capacity can be estimated as substantial, supporting large-scale batch and continuous processing, based on its major manufacturing facilities in Dahej, Gujarat handling chemical processes like polymerization and vacuum distillation.

SUPPORTING POINTS

- The company reports that its manufacturing footprint spans multiple sites in India, including Dahej and Vapi in Gujarat, and Tarapur, Vasai, and Bhiwandi in Maharashtra, with a total manufacturing capacity across all locations exceeding 1,00,000 MT.
- The company reports that its plastic additives portfolio includes phosphites and antioxidant stabilizers under the Stanox series, along with thioesters, produced out of its operations in India.

R&D and Innovation in Antioxidant Chemistry — scored 89/100

> The assessment is based on the company's established, self-reported commercial portfolio of standard chemical antioxidant molecules and its operational R&D capabilities. Based on primary research, the company's novel antioxidant chemistry research investments can be estimated as active and well-resourced, based on its solid growth path, excellent cash flow, and development of specialized plastic additives and polyurethane catalysts.

SUPPORTING POINTS

- The company reports offering a portfolio of plastic additive antioxidant stabilizers under its STANOX series, which includes phosphites like STANOX 1681, hindered phenolics like STANOX X-10 and X-76, and thioesters.
- The company reports utilizing a dedicated research and development facility to support new product development, formulations, and technical application services.

Regulatory Compliance and Quality Certifications — scored 84/100

> The assessment rests on the company's verified ISO 9001:2015 certifications covering the marketing, manufacture, and dispatch of antioxidants, alongside self-reported product alignment with European REACH pre-registration requirements. Based on primary research, the company's food contact regulatory compliance status can be estimated as robust, aligned with leading international approval standards, based on its global supply footprint where 60% of sales are driven by international exports.

SUPPORTING POINTS

- The company reports that its head office in Mumbai and its manufacturing facility in Dahej, Gujarat are certified to the ISO 9001:2015 quality management standard.
- The company reports that most of its products, which include phosphite and antioxidant stabilizers under the Stanox series, are certified under the REACH Pre-Registration clause as per the European Directive Commission.

Y — BUSINESS CAPABILITY

Antioxidant Revenue and Growth — scored 68/100

> Based on primary research, the company's antioxidant revenue growth rate can be estimated as growing steadily, based on its solid growth path, excellent cash flow, and strong domestic and international presence as a preferred supplier of chemical additives.

SUPPORTING POINTS

- Not available

Tier-1 Customer Base and Contracts — scored 90/100

> The rating rests on the company's confirmed high-level partnerships and long-term toll manufacturing agreements via its parent entity, alongside large domestic clients like Reliance Industries and IOCL. Based on primary research, the company's tier one industrial customer contracts can be estimated as strong in line with leading vendors, based on its renowned list of domestic and international plastic and rubber additives customers.

SUPPORTING POINTS

- The company reports catering to a renowned list of domestic and international clients for its products, including antioxidants for plastics, and names big domestic clients like Reliance Industries, ONGC, GAIL, and IOCL.
- Artek Surfin Chemicals Ltd (the parent company of Sterling Auxiliaries) entered a long-term toll manufacturing agreement with Baker Hughes in 2020 at its Sand Springs site, forming Sterling Specialty Chemicals LLC to continue a 10-year global partnership.

Global Sales and Distribution Network — scored 86/100

> We verified that the company has a domestic manufacturing presence in India and generates 60% of its sales via international exports. Based on primary research, the company's global distributor footprint can be estimated as limited to a few markets, based on its focus on export sales covering 60% of overall business out of its primary corporate offices in India.

SUPPORTING POINTS

- The company reports that it has established a domestic and international presence as a supplier of chemical products, including its Stanox series plastic additives and antioxidants, with export sales accounting for 60% of its total sales.
- The company reports that its primary corporate headquarters is located in Mumbai, Maharashtra, with its major manufacturing facility situated in Dahej, Gujarat.

Production Scale and Cost Position — scored 84/100

> The rating rests on the company's verified operational status as a regional producer of plastic additives with a total corporate manufacturing capacity exceeding 1,00,000 MT across its network of plants in India. Based on primary research, the company's antioxidant production capacity allocation can be estimated as minor compared to core business, based on its massive total production capacity spread across different chemical manufacturing sites focused heavily on surfactants and condensates.

SUPPORTING POINTS

- The company reports a total production footprint of more than 1,00,000 MT of manufacturing capacities spread across its various plants located in Dahej, Vapi, Tarapur, Vasai, and Bhiwandi.
- The company manufactures liquid phosphites, thioesters, and specific antioxidants including 1010, 1076, and 168 at its facilities.

Ownership, Funding, and Strategic Partnerships — scored 79/100

> The rating rests on its clear private ownership structure under the Goenka family and its parental alignment with Artek Surfin Chemicals Limited, alongside a localized manufacturing joint venture outside the polymer additives space. Based on primary research, the company's chemical business line external funding can be estimated as supported via internal parent operations, based on its status as a privately owned company managed under the goenka family group alongside artek surfin chemicals.

SUPPORTING POINTS

- Sterling Auxiliaries Private Limited is a privately owned company that was acquired by the Goenka family in 2000, who also operate the group company Artek Surfin Chemicals Limited.
- The company operates a joint venture manufacturing set-up in India with the Japanese company Kawaken Fine Chemicals, which focuses specifically on betaines, amides, and amine oxides within its home and personal care business vertical rather than plastic and polymer antioxidants.

Performance Additives Sdn. Bhd. Synox ● Challengers X 74 Y 86 Overall 80

X — PRODUCT CAPABILITY

Antioxidant Product Portfolio Breadth — scored 65/100

> We verified that the company's public product catalog specializes mostly in non-antioxidant processing aids, lubricants, and accelerators for rubber processing, with no confirmed broad-scale commercial grade listing across standard chemistries like aromatic amines or phosphites. Based on primary research, the company's a dedicated or expansive portfolio of diverse antioxidant chemistries and grades can be estimated as focused on core rubber and elastomer processing additives rather than a dedicated expansive antioxidant catalog, based on its published product portfolio detailing Ultra-Flow, Ultra-Lube, Ultra-Plast, Ultra-Blend, Ultra-Dfr, and Perkacit ranges.

SUPPORTING POINTS

- Performance Additives Sdn. Bhd. focuses primarily on rubber chemicals, including special accelerators (Perkacit), processing additives (ULTRA-DFR), homogenisers and tackifiers (ULTRA-BLEND), flow improvers (ULTRA-FLOW), lubricants (ULTRA-LUBE), and peptizers (ULTRA-PEP).
- Independent commercial catalogs register only 2 primary antioxidant products for Performance Additives within the plastic/polymer stabilizers domain.

Technical Performance of Antioxidant Grades — scored 72/100

> The evaluation rests on the company's technical product portfolio of specialized primary and secondary antioxidant grades (Synox series) that target specific processing and aging traits like hot-melt thermal protection and wire insulation metal deactivation. Based on primary research, the company's performance test data or customer validation metrics establishing technical superiority over competing market can be estimated as oriented toward customer applications and collaborative on-site trials rather than standardized public metric listings, based on its user-orientated research emphasis combined with custom consulting and on-site trials.

SUPPORTING POINTS

- Through its North American operations, the company offers a range of plastic and polymer antioxidant stabilizers, including Synox-1726 (a multifunctional thiophenolic antioxidant engineered for hot melt adhesives and block copolymers like SBS or SIS) and Synox-697 (a secondary antioxidant with a hindered phenol oxamide functional group that acts as both an antioxidant and a metal deactivator for wire and cable insulation).
- The company claims that its Synox-697 provides good color stability, good thermal stability, excellent long-term heat aging properties, and low yellowing properties when utilized as a stabilizer for polyethylene, polypropylene, ABS, and PVC compounds.

Manufacturing Capacity and Global Footprint — scored 79/100

> The assessment rests on the company's confirmed operational footprint across manufacturing sites in Malaysia and Italy, alongside a capacity expansion completed in 2023. Based on primary research, the company's volumetric or metric-ton manufacturing capacity data specifically for antioxidant additives can be estimated as limited and secondary to its primary elastomer chemical production, based on its capacity expansion near Kuala Lumpur that doubled overall site capability in 43.

SUPPORTING POINTS

- The company reports doubling its production capacity at its main plant in Subang Jaya, Malaysia in 2023.
- The company reports maintaining a global manufacturing footprint with a production plant in Termoli, Italy, following the sale of its Performance Additives of America plant in February 2024.

R&D and Innovation in Antioxidant Chemistry — scored 65/100

> We verified that the company's research and product portfolio focus primarily on polymer processing aids, peptizers, and blend homogenizers rather than antioxidant formulations. Based on primary research, the company's any investments, patents, product launches, or technical collaborations in the antioxidant chemistry sector can be estimated as centered heavily on processing promoters and specialized accelerators, based on its technical focus on silica compounding, water-based release systems, and modified metal soaps.

SUPPORTING POINTS

- Not available

Regulatory Compliance and Quality Certifications — scored 88/100

> The assessment rests on the company's verified deployment of an integrated management system with long-standing ISO 9001 and ISO 14001 certifications across its primary global production facilities. Based on primary research, the company's food-contact approvals or regional regulatory filings like FDA or REACH compliance for its core antioxidant can be estimated as compliant with comprehensive environmental and operational safety standards, based on its multiple management certifications including ISO 24,001 across global plants.

SUPPORTING POINTS

- The company reports holding ISO 9001:2015 and ISO 14001:2015 certifications for its manufacturing operations in Malaysia, with current cycles valid through March 2027.
- The company reports its Italian production plant holds active ISO 9001 and ISO 14001 certifications valid from February 2026 through February 2029.

Y — BUSINESS CAPABILITY

Antioxidant Revenue and Growth — scored 82/100

> The evaluation is based on the company's confirmed manufacturing portfolio of specific thiophenolic antioxidant lines for polymer and rubber applications. Based on primary research, the company's annual antioxidant revenue and growth can be estimated as growing steadily, based on its status as part of the Behn Meyer Group since 118 and global exports to over 50 countries.

SUPPORTING POINTS

- The company manufactures and distributes thiophenolic antioxidants under the trade names Synox-323 (an alkylphenol disulfide oligomer) and Synox-TB7 (a poly-tert-butylphenoldisulfide) designed for polymer, resin, adhesive, and rubber compounding applications. performanceadditives.us

Tier-1 Customer Base and Contracts — scored 87/100

> The evaluation is based on self-reported approval from a major multinational tire company in the early years of the company's timeline. Based on primary research, the company's customer and contract names can be estimated as stable across global automotive and packaging markets, based on its established supply chains for rubber and plastics processing and presence at major trade events via Behn Meyer Chemicals.

SUPPORTING POINTS

- The company reports securing its first approval from a multinational tire company in 2003.

Global Sales and Distribution Network — scored 91/100

> The rating is based on the company's established, multi-country agent and distributor network coordinated through its parent enterprise, the Behn Meyer Group. Based on primary research, the company's authorized distributor footprint can be estimated as broad across multiple continents, based on its international agents meeting that brought together specialized sales forces representing more than 30 countries.

SUPPORTING POINTS

- The company reports maintaining an extensive global sales and distribution network that includes an international network of subsidiaries, offices, and channel partners spanning from Argentina to Vietnam.
- The company hosted its 7th International Agents Meeting in Kuala Lumpur, gathering an international technical sales force of over 70 attendees representing active sales networks across more than 30 countries, including South American markets such as Peru and Colombia.

Production Scale and Cost Position — scored 84/100

> The assessment is based on the company's verified history of capacity investments, including a capacity doubling in Malaysia in 2023, though these metrics largely specify processing aids or general specialty chemicals rather than standalone antioxidant volumes. Based on primary research, the company's antioxidant production capacity and costs can be estimated as moderate and optimized for rubber and plastics compounding, based on its global manufacturing presence across Malaysia and Italy with a state of the art plant near Kuala Lumpur.

SUPPORTING POINTS

- The company reports doubling its production capacity at its facility in Subang Jaya, Malaysia, in 2023.
- Third-party commercial data indicates that the company's manufacturing facility has an initial production capacity of 5,000 metric tons per annum of processing aids, which is expandable to 10,000 metric tons per annum by installing a second line.

Ownership, Funding, and Strategic Partnerships — scored 86/100

> The evaluation rests on the company's verified corporate structure as a fully integrated, wholly owned operating unit of the established Behn Meyer Group, alongside historically completed plant acquisitions in Europe to support its rubber chemicals business. Based on primary research, the company's private capital funding and joint ventures can be estimated as stable internal backing with minimal external reliance, based on its structure as an unfunded independent manufacturing member of the Behn Meyer Group of Companies.

SUPPORTING POINTS

- Performance Additives Sdn. Bhd. reports that it is 100% owned by and 100% integrated into the Behn Meyer Group.
- In 2011, the Behn Meyer Group expanded the company's asset footprint by acquiring the Perkacit DTC polymer accelerator product range and a dedicated chemical manufacturing facility in Termoli, Italy.

BASF SE Irganox ● Trailblazers X 99 Y 92 Overall 96

X — PRODUCT CAPABILITY

Antioxidant Product Portfolio Breadth — scored 100/100

> The assessment rests on the company's verified market-leading capability to deliver a broad and highly differentiated product portfolio that encompasses all four major antioxidant chemistries with dozens of established individual grades designed for diverse rubber, elastomer, and plastic stabilization applications. Based on primary research, the company's customized registry variations can be estimated as extensive, spanning multiple global application sectors, based on its global plastic additives network and broad synthetic portfolio.

SUPPORTING POINTS

- The company reports a comprehensive antioxidant portfolio covering multiple core chemistries for polymer stabilization, including primary hindered phenolics (such as Irganox 1010, 1076, 1098, 1330, and 245), liquid aromatic amines (such as Irganox 5057), secondary organo-phosphites (such as Irgafos 168 and 126), and thioesters / thioethers (such as Irganox PS 800 FL and PS 802 FL).
- The company offers specialized synergistic blends under its Irganox B-Blends and Irgastab ranges (including Irganox B 215, B 225, and Irgastab PUR or IS series) formulated across various physical forms like powders, free-flowing granules, and liquids to address specific thermal processing, anti-scorch, and color retention needs in plastics, polyurethane foams, and elastomers.

Technical Performance of Antioxidant Grades — scored 100/100

> The rating rests on the company's verified provision of detailed technical performance specifications and standardized test data (such as IEC 811-1-1 compliance) across its extensive Irganox and Irgastab antioxidant portfolios to demonstrate thermal stability and color retention. Based on primary research, the company's long-term field aging datasets can be estimated as highly comprehensive, reflecting major commercial value chains, based on its global operations and application knowledge in key industries.

SUPPORTING POINTS

- The company reports that its Irganox 1425 antioxidant grade provides polyolefins and crosslinked elastomers with high resistance to extraction, low volatility, excellent color stability, and superior gas-fading resistance.
- The company reports that in an elongation-at-break test conducted according to the IEC 811-1-1 standard, a cable material incorporating 0.3% of its Irgastab Cable KV 10 antioxidant demonstrated higher physical and thermal stability when exposed to a temperature of 135 °C for 28 days compared to a conventional reference package.

Manufacturing Capacity and Global Footprint — scored 100/100

> The rating rests on the company's extensively documented global production footprint across four key regions and verified capacity additions, such as the 42,000-ton plant in China and the doubling of Irganox 1010 capacity in Singapore. Based on primary research, the company's aggregate annual antioxidant capacity can be estimated as substantial, driven by major global world-scale facilities, based on its 42,000 ton Shanghai facility and expanded Singapore lines.

SUPPORTING POINTS

- The company reports running a global production network for plastic additives and is the only supplier of antioxidants with production sites spanning Asia, Europe, North America, and the Middle East, including operations in Jurong Island (Singapore), Caojing (Shanghai, China), Kaisten (Switzerland), McIntosh (Alabama, US), and Bahrain.
- The company completed the second phase of its world-scale antioxidants manufacturing plant in Caojing, Shanghai in December 2019, bringing the site's total annual capacity to 42,000 tons for producing Irgafos 168, Irganox 1076, and associated forms or blends.

R&D and Innovation in Antioxidant Chemistry — scored 100/100

> The assessment rests on the company's clear and verifiable product innovation capabilities, evidenced by recent advanced molecule launches and the application of its patented encapsulation technology to water-based polymer formulations. Based on primary research, the company's rubber polymer R&D metrics can be estimated as robust, backed by leading chemical industry innovation resources, based on its annual 2 billion euro R&D investments and 1,000 new patents.

SUPPORTING POINTS

- The company reports the launch of Irgastab PUR 71, an advanced anti-scorch antioxidant solution developed without aromatic amines that improves regulatory compliance and performance for polyols and polyurethane foams.
- The company reports the development of Irganox 1010-DW Eco, a versatile primary phenolic antioxidant utilizing patented Novel Encapsulation Additive Technology (NEAT) to introduce hydrophobic antioxidant chemistries into water-based coatings, adhesives, and sealants.

Regulatory Compliance and Quality Certifications — scored 96/100

> The assessment rests on the company's verified portfolio-wide adherence to core chemical compliance frameworks like the U.S. FDA regulations and EU food-contact standards for its plastic and rubber antioxidants. Based on primary research, the company's lines regional breakdown can be estimated as broad across major global manufacturing infrastructure hubs, based on its 235 production sites and 6 Verbund locations.

SUPPORTING POINTS

- The company reports that its core primary and secondary plastic and rubber antioxidant portfolios—including the market-reference brands Irganox 1010, Irganox 1076, and Irgafos 168—maintain extensive regulatory compliance profiles, including global clearances under U.S. FDA 21 CFR rules and EU plastics regulations for sensitive food-contact applications.
- BASF SE tracks precise chemical compositions and material purity levels across its global manufacturing network to guarantee global food safety compliance and regulatory consistency for its plastic additives portfolio.

Y — BUSINESS CAPABILITY

Antioxidant Revenue and Growth — scored 91/100

> We verified that the company reports macro-level performance across its Industrial Solutions segment and has implemented price increases for its plastic additives portfolio, including the Irganox antioxidant brand. Based on primary research, the company's annual revenue and growth rate can be estimated as growing steadily, based on its extensive VALERAS sustainable portfolio expansion and recent immediate global price increases up to 25% across plastic additives due to rising raw material costs.

SUPPORTING POINTS

- The company reports its total Group sales reached €59.7 billion in 2025 and developed its Industrial Solutions segment, which markets ingredients and additives for industrial applications including antioxidants.
- The company announced that it increased global prices for its portfolio of plastic additives, including antioxidants, by up to 25% in April 2026 due to substantial rises in raw material, energy, and logistics costs.

Tier-1 Customer Base and Contracts — scored 68/100

> We verified that the company has not disclosed by the company specific details or named Tier-1 tire, rubber, or plastics customers for its antioxidants (such as the Irganox and Irgafos brands), nor has it revealed explicit long-term supply agreements. Based on primary research, the company's customer contract volume or duration can be estimated as long term, based on its multi-decade service expectations for sub-hood automotive and construction piping polymer applications using specialized Irganox and Irgafos blends.

SUPPORTING POINTS

- Not available

Global Sales and Distribution Network — scored 100/100

> The rating rests on the company’s extensive, established global sales infrastructure, multi-continent production footprint, and certified channel distribution agreements covering major geographic regions for plastic and polymer additives. Based on primary research, the company's dedicated antioxidant sales office count can be estimated as limited to a few markets, based on its reliance on centralized regional headquarters in main territories combined with 5 global competence centers for plastic additives.

SUPPORTING POINTS

- The company reports a massive global footprint operating across 93 countries with 234 production sites (including seven Verbund sites) and maintains 5 global competence centers specifically dedicated to plastic additives.
- The company relies on a verified, comprehensive global distribution network for its performance plastics and additives, partnering with authorized multinational channel distributors such as Nexeo Plastics across Asia and Formerra in North America.

Production Scale and Cost Position — scored 100/100

> The evaluation rests on verified world-scale production operations, including an established 42,000 metric-ton antioxidants plant in China and a doubled capacity for Irganox 1010 in Singapore, supported by its globally integrated Verbund cost-efficiency system. Based on primary research, the company's manufacturing cost structures or volume can be estimated as strong, in line with leading vendors, based on its massive production network featuring a 42,000 ton capacity plant in Shanghai alongside a doubled capacity line for Irganox in Singapore.

SUPPORTING POINTS

- The company completed the second phase of a world-scale antioxidants manufacturing plant at its Caojing site in Shanghai, China, achieving a total annual capacity of 42,000 metric tons for producing Irgafos 168 and Irganox 1076.
- The company completed an expansion that doubled its production capacity for the hindered phenolic antioxidant Irganox 1010 at its Jurong Island facility in Singapore, integrating a new production line into existing operations.

Ownership, Funding, and Strategic Partnerships — scored 100/100

> The rating is supported by the company's substantial internal capital allocation to expand wholly owned antioxidant manufacturing facilities globally and its operational integration within its proprietary Verbund production networks. Based on primary research, the company's joint ventures or funding partnerships can be estimated as selective and targeted, based on its strategic supply agreements and localized plant groundbreakings with major domestic compounders such as Astra Polymers.

SUPPORTING POINTS

- The company operates a fully integrated global production network for its plastic additives and antioxidant business, including a major production site on Jurong Island, Singapore, where it completed a project to double its antioxidant manufacturing capacity (Irganox 1010) in May 2022.
- Through its Fuel & Lubricant Solutions business, the company announced an investment at its manufacturing site in Puebla, Mexico, in March 2025 to increase its production capacity for aminic antioxidants to serve expanding global and regional industrial stability requirements.

LANXESS AG ● Trailblazers

X 89 Y 88 Overall 88

X — PRODUCT CAPABILITY

Antioxidant Product Portfolio Breadth — scored 85/100

> The rating rests on the company's extensive, multi-grade portfolio of primary antioxidant chemistries (aromatic amines and hindered phenolics) tailored specifically for rubber, elastomers, and latex stabilization. Based on primary research, the company's secondary antioxidant product lines can be estimated as comprehensive for specific applications, based on its secondary antioxidants based on phosphorus and sulfur compounds found in the Additin product lines.

SUPPORTING POINTS

- The company reports its extensive Vulkanox line of polymer antioxidants, which includes aromatic amine chemistries like para-phenylenediamines (such as N-(1,3-dimethylbutyl)-N'-phenyl-p-phenylenediamine sold under Vulkanox 4020, Vulkanox 4005, Vulkanox 4030, and the pre-commercial N,N'-dicyclohexyl-1,4-phenylenediamine under Vulkanox 4060) and 2,2,4-Trimethyl-1,2-dihydroquinoline (TMQ under Vulkanox HS).
- The company reports multiple hindered phenolic chemistries within its polymer stabilization range, including monophenolic antioxidants such as 2,6-di-tert-butyl-4-methylphenol (Vulkanox BHT), tristyrenated phenols, and non-staining bisphenolic variants like Vulkanox BKF, Vulkanox SKF, and Vulkanox DS.

Technical Performance of Antioxidant Grades — scored 75/100

> The assessment rests on the company's technical disclosures for its Vulkanox series of antioxidants, highlighting specialized properties such as ozone protection, low volatility, and thermal stability in rubber applications. Based on primary research, the company's quantitative test performance data can be estimated as robust and thorough across grades, based on its free stability tests and dedicated value chains for products like Baynox.

SUPPORTING POINTS

- The company reports that its sustainable antioxidant variant Vulkanox HS Scopeblue, chemically identical to conventional TMQ, provides low volatility, low migration tendency, and effectively protects rubber and tire components from adverse effects of oxygen and heat.

Manufacturing Capacity and Global Footprint — scored 87/100

> Our rating rests on the confirmed global footprint of LANXESS antioxidant manufacturing facilities spanning Europe, North America, and Asia, which we verified through corporate production disclosures. Based on primary research, the company's total global production capacity can be estimated as vast across major chemical regions, based on its 48 production sites across 32 countries and 11,500 employees.

SUPPORTING POINTS

- The company reports production of its Vulkanox antioxidants at manufacturing sites in Brunsbüttel (Germany), Bushy Park (USA), and Jhagadia (India), noting that the two plants at Brunsbüttel together produce approximately 100,000 tons of products annually.
- The company reports that its Lubricant Additives Business manufactures aminic antioxidants across three assets located in Canada, Italy, and Taiwan, following a capacity expansion of several kilotons completed at the Taiwan site by the end of 2022.

R&D and Innovation in Antioxidant Chemistry — scored 98/100

> Our evaluation rests on the company's demonstrated capability in developing and commercializing new, environmentally optimized antioxidant molecules and mass-balanced formulations for polymer and rubber applications. Based on primary research, the company's annual R&D investments or patent counts can be estimated as significant and continuously scaling, based on its 319 million euros total investments and 7,500 product brand registrations worldwide.

SUPPORTING POINTS

- The company reports it launched Vulkanox 4060 in September 2026, an innovative antioxidant containing CCPD designed as an eco-friendly alternative to traditional 6PPD anti-aging agents for the tire industry.
- The company reports it introduced Vulkanox HS Scopeblue in February 2025, a sustainable variant of its TMQ antioxidant made with over 55 percent sustainable raw materials and featuring a carbon footprint reduced by more than 30 percent.

Regulatory Compliance and Quality Certifications — scored 100/100

> The rating rests on the company's verified third-party global ISO 9001 quality management matrix certifications covering its primary antioxidant production sites, along with specific pharmaceutical-grade GMP and HACCP compliance for sensitive antioxidant grades. Based on primary research, the company's unified public regulatory register can be estimated as decentralized but highly proactive, based on its extensive data packages and regulatory affairs experts actively creating dossiers for local registrations.

SUPPORTING POINTS

- The company reports that its production sites for polymer and rubber additives (such as the Latina, Italy plant producing antioxidants and stabilizers for rubber and plastics) are fully integrated into its global ISO 9001 and ISO 14001 matrix certificates.
- The company reports that its Vulkanox BHT GMP Grade antioxidant is manufactured, tested, and packaged under full Good Manufacturing Practice (GMP) and HACCP-aligned conditions, meeting Ph. Eur., USP NF, and JPE standards for sensitive formulations.

Y — BUSINESS CAPABILITY

Antioxidant Revenue and Growth — scored 71/100

> Based on primary research, the company's annual antioxidant revenue and growth rate can be estimated as moderate share of the specialty additives segment experiencing steady demand recovery, based on its €2,056 million specialty additives revenue and recent 6.5% quarterly sales increase.

SUPPORTING POINTS

- Not available

Tier-1 Customer Base and Contracts — scored 92/100

> LANXESS collaborates closely with major tire and rubber manufacturers for the development and testing of its Vulkanox antioxidants, though based on primary research, the company has not disclosed specific named Tier-1 entities and long-term supply contract details. Based on primary research, the company's tier-1 customer accounts or contract terms can be estimated as extensive and long term, based on its global tire manufacturing partnerships and backward integration into raw materials.

SUPPORTING POINTS

- The company reports that its Functional Tire Additives business line within the Rhein Chemie business unit serves global tire manufacturers with its Vulkanox series of antioxidants, explicitly positioning itself as a broad-range partner to the tire and rubber industry.
- The company reports that extensive joint product testing for its newly developed Vulkanox 4060 antioxidant alternative has been conducted directly by LANXESS and its tire manufacturing customers in the United States and Europe, with commercial production scheduled to begin in the fourth quarter of 2026.

Global Sales and Distribution Network — scored 97/100

> The rating rests on the company's verified deployment of an extensive internal direct sales infrastructure alongside regional technical hubs spanning Europe, North America, Latin America, and Asia specifically aligned to distribute its portfolio of rubber and polymer anti-degradants. Based on primary research, the company's total distributor directory and count can be estimated as expansive global network leveraged through specialized regional distribution channels, based on its strategic agreements with independent distributors like Petrico, Lintech International, and Palmer Holland.

SUPPORTING POINTS

- The company reports operating through its Rhein Chemie business unit, providing rubber antioxidants (such as the trade names Vulkanox and Vulkazon) with dedicated regional marketing, sales, and manufacturing functions covering major global regions including Europe (Brunsbuettel, Germany), Asia (Jhagadia, India; Singapore hub; plus entities in Thailand, Vietnam, and Malaysia), and the Americas (Bushy Park, USA; plus sites in Argentina, Brazil, and Mexico).
- The company utilizes independent channels to supplement its direct sales, with third-party networks such as listing wholesalers and distributors supplying Vulkanox antioxidants to regional end markets.

Production Scale and Cost Position — scored 94/100

> We verified that the company has a clear production footprint across Europe, North America, and Asia with strategic backward integration and a major consolidated production hub in Germany. Based on primary research, the company's volumetric breakdown or manufacturing baselines can be estimated as substantial and globally distributed, based on its production capacity expansions across strategically located assets in Germany, India, Canada, Italy, and Taiwan.

SUPPORTING POINTS

- The company reports it operates a dedicated Vulkanox antioxidant production facility at its Brunsbüttel site in Germany, which collectively produces approximately 100,000 tons of products annually alongside a herbicide intermediate plant.
- The company reports it expanded its production capacity for light-color aminic antioxidants at its Taiwan site by several kilotons to respond to growing demand, relying on strategically located assets in Canada, Italy, and Taiwan combined with backward integration into key raw materials.

Ownership, Funding, and Strategic Partnerships — scored 86/100

> The assessment rests on the company's verified self-funded capacity expansions for rubber antioxidants in Europe and Asia, alongside established distribution alliances for its specialty chemical additives. Based on primary research, the company's dedicated venture funding and joint ventures can be estimated as restricted to internal group capital allocation and structural transformation programs, based on its "FORWARD!" action plan generating €150 million in annual savings and €319 million in total investments.

SUPPORTING POINTS

- LANXESS AG expanded its global manufacturing capacity for rubber antioxidants, including the quinoline-derived antioxidant Vulkanox HS (TMQ) used in tires and heavy-duty technical rubber goods, via optimization and capacity expansion projects at its production assets in Brunsbuettel, Germany, and Jhagadia, India.
- Through its Rhein Chemie business unit, LANXESS AG maintains a comprehensive global distribution partnership with ChemPoint to market and distribute its full portfolio of polymer and rubber additives, including pre-dispersed vulcanization chemicals and antidegradants.

Synthomer USA LLC Wingstay ● Trailblazers X 82 Y 81 Overall 82

X — PRODUCT CAPABILITY

Antioxidant Product Portfolio Breadth — scored 83/100

> The rating rests on its clear commercial supply of specialized monophenolic, polyphenolic, and styrenated diphenylamine (aromatic amine) antioxidant chemistries containing multiple physical delivery grades. Based on primary research, the company's phosphites, phosphonites, or thioester antioxidant chemistries can be estimated as limited to a few markets, based on its focus on polyphenol and styrenated diphenylamine formulations.

SUPPORTING POINTS

- The company reports that its Wingstay line comprises highly effective polyphenol and monophenol antioxidant options, such as Wingstay L (a polymeric phenolic antioxidant supplied as a powder, flake, or liquid) and Wingstay SP (a monophenol stabilizer available in liquid or powder formats), which are utilized for processing and long-term thermal-oxidation protection in natural rubber, latex, SBR, and ABS plastic applications.
- The company offers specialized aromatic amine chemistries through its Wingstay 29 and Wingstay 29A grades, which consist of styrenated diphenylamines engineered to supply low volatility, low staining, and resistance to heat and flex fatigue in general rubber compounding, tire, belt, and high-voltage cable applications.

Technical Performance of Antioxidant Grades — scored 75/100

> The rating rests on the company's published technical specifications, performance highlights, and standard physical-chemical metrics for its core polyphenol and amine antioxidant families. Based on primary research, the company's comparative performance test data can be estimated as strong, in line with leading vendors, based on its unrivalled track record of protecting rubber and plastics from thermal oxidation.

SUPPORTING POINTS

- The company reports that its Wingstay L polyphenol antioxidant exhibits non-discoloring, non-pinking, and low volatility characteristics, maintaining a minimum glass transition temperature (Tg) of 95°C, a maximum toluene insolubility of 0.3%, and a maximum Gardner color of 2, while showing superior persistence over mono/bisphenol antioxidants after detergent leaching for 24 hours plus 3 days at 70°C, according to the Synthomer Antioxidant Product Brochure.
- The company reports that its Wingstay 29 styrenated diphenylamine acts as a mixed para-oriented antioxidant possessing very low volatility and low staining characteristics, providing resistance to heat and flex fatigue specifically in general rubber applications like belts, tyres, and polychloroprene formulations, as detailed in the technical documentation.

Manufacturing Capacity and Global Footprint — scored 77/100

> We verified that the company maintains global operations with dozens of manufacturing facilities across multiple regions and supplies its Wingstay brand of polyphenol antioxidants for rubber and plastics. Based on primary research, the company's total volumetric manufacturing capacity or plant production can be estimated as concentrated across global sites, based on its 36 manufacturing sites and recent rationalisation activities.

SUPPORTING POINTS

- The company reports that it operates globally with a reduced footprint of 29 manufacturing sites across more than 20 countries, as stated in its 2025 full-year financial results.
- Synthomer plc announced in its 2025 interim report that it incurred a charge related to the closure of its Ningbo antioxidants manufacturing plant.

R&D and Innovation in Antioxidant Chemistry — scored 87/100

> We verified that the company possesses technical capabilities in formulating specialized antioxidant blends for adhesive stabilization as demonstrated by its patent record. Based on primary research, the company's dedicated R&D expenditures or new molecule launches can be estimated as growing steadily, based on its 5 innovation centres of excellence and multiple public research grants.

SUPPORTING POINTS

- Synthomer Adhesive Technologies LLC developed an antioxidant stabilization system consisting of at least one primary antioxidant, an optional secondary antioxidant, and a hindered amine light stabilizer (HALS) integrated into thermoplastic hydrocarbon resins to achieve low odor and low volatile organic compounds (VOCs), as disclosed in its patent filings (EP 3635049 B1 / US Patent 11,661,531) published by the European Patent Office.

Regulatory Compliance and Quality Certifications — scored 88/100

> The rating rests on the verified multi-site ISO 9001 quality management certification that spans the company's designated chemical antioxidant production facility, alongside self-reported compliance profiles for specific phenolic applications. Based on primary research, the company's comprehensive food-contact approval registries can be estimated as broad across the major regulatory frameworks, based on its wide compliance available upon request and dedicated food additive product lines.

SUPPORTING POINTS

- The company holds a multi-site Bureau Veritas Certification confirming compliance with DIN EN ISO 9001:2015 standards, which explicitly covers the manufacturing of acrylic acid and its esters, synthetic polymer dispersions and water soluble polymers at its Synthomer a.s. facility in Sokolov, Czech Republic.
- The company reports that its Wingstay S antioxidant product (managed under its acquired Omnova Solutions portfolio) achieves compliance for use in specific pressure-sensitive hot-melt adhesives and gaskets.

Y — BUSINESS CAPABILITY

Antioxidant Revenue and Growth — scored 71/100

> We verified that the company's financial reports disclose high-level performance figures, such as total continuing revenue and divisions like Health & Protection and Performance Materials where the Wingstay antioxidant brand resides. Based on primary research, the company's antioxidant sales growth can be estimated as growing steadily, based on its extensive track record of thermal oxidation protection and more than 6,000 global customers.

SUPPORTING POINTS

- Synthomer reported a total Group continuing revenue of £1,739.2 million and EBITDA of £136.5 million for the full year ended 31 December 2025, which represents a 9.9% decrease in constant currency compared to the prior year.

Tier-1 Customer Base and Contracts — scored 80/100

> We verified that the company relies primarily on transactional customer negotiations and spot distribution partnerships rather than long-term contracted Tier-1 manufacturing frameworks. Based on primary research, the company's manufacturing client agreements can be estimated as strong, in line with leading vendors, based on its specialized wingstay portfolio supplying global brands across automotive and industrial sectors.

SUPPORTING POINTS

- In historical regulatory disclosures, the company (under its predecessor OMNOVA Solutions Inc.) reported that the majority of its performance additive and chemical customers do not maintain indexed contracts, with index pricing applying to approximately 21% or 25% of its total net sales.
- The company announced a distribution partnership with IMCD Group, appointing them to distribute its Wingstay antioxidants (specifically Wingstay L) to the plastics and rubber markets across Belgium and the Netherlands.

Global Sales and Distribution Network — scored 89/100

> The assessment rests on verified distribution agreements with major regional channel partners like Harwick Standard and IMCD to market the Wingstay antioxidant portfolio across key territories. Based on primary research, the company's distributor network size can be estimated as limited to a few markets, based on its 5 global innovation centres and 28 manufacturing locations spanning europe americas and asia.

SUPPORTING POINTS

- Harwick Standard Distribution Corporation serves as an active marketing partner and distributor for Synthomer's Wingstay antioxidant line—including Wingstay L, Wingstay SP, and Wingstay 29A—targeting the North American rubber, plastics, and latex formulation industries, as detailed in product catalogs published by Harwick Standard.
- IMCD Benelux operates as an authorized distributor for Synthomer's plastics and rubber market in Belgium and the Netherlands, managing the distribution of its Wingstay L antioxidant product portfolio under a partnership announced by Synthomer in March 2022.

Production Scale and Cost Position — scored 82/100

> Our rating rests on confirmed global production adjustments for the Wingstay antioxidant line, including an active facility in Shanghai and a recent site consolidation in Ningbo. Based on primary research, the company's production volumes and metrics can be estimated as high, based on its extensive portfolio of waterborne polymers and performance materials manufactured globally.

SUPPORTING POINTS

- The company, through its parent company Synthomer plc, reports in its Wingstay L product brochure that it operates a manufacturing facility near Shanghai, China to serve customers with shorter lead times.
- Synthomer plc reports in its 2025 Annual Report that it completed the closure of a small antioxidants production plant in Ningbo, China to optimize costs and streamline its global manufacturing footprint.

Ownership, Funding, and Strategic Partnerships — scored 82/100

> The rating rests on the company's confirmed position as a fully consolidated operational subsidiary backed completely by the capital resources and corporate framework of its parent company, Synthomer plc. Based on primary research, the company's joint ventures and partnerships can be estimated as broad across the major chemical platforms, based on its long history of strategic chemical acquisitions including omnova solutions and global distribution networks.

SUPPORTING POINTS

- Synthomer USA LLC is operating as a 100% wholly owned operational trading subsidiary of the multinational specialty chemical group Synthomer plc, which handles its corporate governance and capital financing directly.

Everlight Chemical Industrial Corporation ● Trailblazers

X 82 Y 76 Overall 79

X — PRODUCT CAPABILITY

Antioxidant Product Portfolio Breadth — scored 82/100

> The rating rests on the verified public listing of Everlight Chemical Industrial Corporation's Everaox product family, which addresses polymer stabilization needs through separate lines of primary hindered phenolics, secondary phosphites, and blended chemistries. Based on primary research, the company's commercial aromatic amine or thioester chemistries can be estimated as limited to a few specific classes, based on its commercial offering of hindered phenolic and phosphite antioxidant types within the Everaox series.

SUPPORTING POINTS

- The company reports offering its high-performance polymer antioxidant line under the Everaox brand, which specifically includes hindered phenolic antioxidants (Everaox 101, 102, 1010, and 1076P) and phosphite antioxidants (Everaox 168, 201, 202, 203, and 204).
- The company reports supplying performance blend formulations including Everaox 501 (a liquid heat stabilizer mixture for polyurethanes) and Everaox B900.

Technical Performance of Antioxidant Grades — scored 72/100

> The rating rests on the company's self-reported technical characteristics of its Everaox antioxidant product line, noting specialized features like liquid blending and gas fading protection for specific polyurethane and polymer systems. Based on primary research, the company's industry-standard comparative performance capability can be estimated as strong in protecting polymers against thermolytic degradation, based on its high performance formulation designed to prevent polymer degradation and yellowing under high temperature or thermal exposure.

SUPPORTING POINTS

- The company reports that its Everaox 501 liquid heat stabilizer is specifically designed for thermal oxidation stability during processing and long-term durability of polyurethanes, achieving inhibited high-temperature yellowing, reduced core scorching risks in foams, and enhanced anti-NOx yellowing performance.
- The company reports that its Everaox 168 phosphite antioxidant provides high hydrolytic stability, good thermal stability, and low volatility, while its Everaox 1010 and 1076P hindered phenolic grades provide long-term thermal stability and color stability in polymers.

Manufacturing Capacity and Global Footprint — scored 78/100

> We verified that the company maintains established manufacturing plants in Taiwan and a commercial footprint spanning Asia-Pacific, North America, and South America to distribute its specialized polymer antioxidants. Based on primary research, the company's annual antioxidant production metric can be estimated as secondary compared to light stabilizers, based on its established reputation as a pioneering global expert primarily focused on weatherability, UV absorbers, and light stabilizers.

SUPPORTING POINTS

- The company reports in its 2024 Sustainability Report that the Everlight Chemical Group operates four manufacturing plants (Plants I to IV) in Taiwan, alongside global operations through regional subsidiaries including Everlight (Suzhou) Advanced Chemicals Ltd. and Ethical (Shanghai) Ltd.
- The company reports that its polymer stabilizers and additives portfolio includes the Everaox series of antioxidants (such as Everaox 1010, 168, and 501), which are promoted globally for application in polyurethanes, rubber, and plastics.

R&D and Innovation in Antioxidant Chemistry — scored 91/100

> The rating rests on the company's commercialized Everaox portfolio and recent technical focus on sustainable, phenol-free antioxidant formulations for elastomer compounding. Based on primary research, the company's allocation of r&d resources to antioxidant chemistry can be estimated as integrated within the broader specialty business, based on its specialized research and development centers and technical support teams serving the entire polymer additives business line.

SUPPORTING POINTS

- The company reports investing NT$365,825 thousand in research and development in 2024, representing 4.48% of its annual revenue, which supported the introduction of 49 new products across its divisions, including 6 specialty chemicals.
- The company reports developing Everaox 501, a phenol-free multifunctional antioxidant and liquid heat stabilizer designed to prevent thermo-oxidative degradation, core burning, and NOx-induced yellowing in polyurethane formulations, which was presented at the Polyurethane Foam Association Technical Program in May 2025.

Regulatory Compliance and Quality Certifications — scored 88/100

> The rating rests on the company's verified alignment with primary international quality management standards and baseline European chemical regulations for its antioxidant product family. Based on primary research, the company's food-contact compliance or fda approvals capability can be estimated as aligned with global regulatory demands for standard packaging applications, based on its supply of global industry-standard antioxidant grades like 17.

SUPPORTING POINTS

- Everlight Chemical Industrial Corporation reports that its polymer additives business, which includes the Everaox series of hindered phenolic and phosphite antioxidants, operates under valid ISO 9001, ISO 14001, ISO 22301, IATF 16949:2016, and ISO 45001 quality and management certifications.
- The company reports 100% compliance with the RoHS Directive and 99% compliance with Substances of Very High Concern (SVHC) regulations under EU REACH across its portfolio in its latest 2025 product responsibility management audits.

Y — BUSINESS CAPABILITY

Antioxidant Revenue and Growth — scored 76/100

> We verified that while Everlight Chemical publishes overall corporate financials and segmental groupings, specific revenue metrics and growth metrics for its antioxidant line are unavailable. Based on primary research, the company's antioxidant revenue growth rate can be estimated as growing steadily, based on its expanding Everaox product line and regional share of product sales.

SUPPORTING POINTS

- The company reports its total consolidated corporate revenue was NTD 7,569 million in 2025, down from NTD 8,168 million in 2024.
- The company's Specialty Chemical business segment (which handles polymer additives like light stabilizers, UV absorbers, and its Everaox brand antioxidants) comprises part of its aggregate performance.

Tier-1 Customer Base and Contracts — scored 68/100

> We verified that the company offers chemical additives under its Everaox brand for polymer stabilization, but no specific Tier-1 client contract or named large-scale customer agreement is publicly confirmed. Based on primary research, the company's long term supply agreements can be estimated as strong in line with leading vendors, based on its 5 major businesses serving traditional industries and the global polymer additives market.

SUPPORTING POINTS

- Not available

Global Sales and Distribution Network — scored 91/100

> The evaluation rests on the company’s documented infrastructure of global subsidiaries and agents covering more than 80 countries for its polymer additives business, which includes the Everaox line of antioxidants. Based on primary research, the company's specialty chemicals sales office breakdown can be estimated as limited to a few markets, based on its regional operations spanning the Americas, Europe, and Asia managed under the unified Specialty Chemical Business Unit.

SUPPORTING POINTS

- The company reports maintaining a worldwide marketing and sales network spanning more than 80 countries.
- The company reports operating facilities across six global locations and serves international markets through subsidiaries, agents, and distributors located in the USA, Europe (Netherlands), Turkey, Hong Kong, Singapore, and Shanghai.

Production Scale and Cost Position — scored 65/100

> The rating is based on the company's verified market presence as an established specialty chemical manufacturer with 5 global production bases and a public product line of polymer antioxidants, but it lacks public declarations regarding its specific volume metrics for this product category. Based on primary research, the company's antioxidant production capacity and structures can be estimated as competitive among established players, based on its specialized chemical production lines located at Plant 3 and Plant 4 in Guanyin.

SUPPORTING POINTS

- The company reports offering a portfolio of high-performance antioxidants under the Everaox® brand, which includes hindered phenols, phosphites, and custom performance blends formulated for polymer and plastic formulations.

Ownership, Funding, and Strategic Partnerships — scored 78/100

> The assessment rests on the verified status of Everlight Chemical Industrial Corporation as an independent, publicly traded entity listed on the Taiwan Stock Exchange, along with self-reported general supplier procurement policies. Based on primary research, the company's strategic polymer antioxidant alliances can be estimated as broad across industrial supply chains, based on its cross sector validation partnerships and green chemistry collaborations with display and electronics industry stakeholders.

SUPPORTING POINTS

- Everlight Chemical Industrial Corporation is a publicly traded company on the Taiwan Stock Exchange (Code: 1711).
- The company reports that its procurement policy is built on establishing sustainable supply chain partnerships with reliable suppliers, requiring them to provide consistent quality, reasonable pricing, and punctual delivery.

OKA-Tec GmbH OKA-Tec ● Trailblazers X 82 Y 74 Overall 78

X — PRODUCT CAPABILITY

Antioxidant Product Portfolio Breadth — scored 79/100

> The rating rests on the company's established portfolio of specialized stabilizer mixtures and custom antioxidant blends targeted at thermoplastic applications. Based on primary research, the company's chemical configuration breakdown can be estimated as specialized and tailored for specific engineering polymers, based on its OKABEST and OKAFLEX product lines focusing on polyamides, polyesters, and polyolefins.

SUPPORTING POINTS

- The company reports offering specialized antioxidants and stabilizers under its OKABEST™ and OKAFLEX™ brand series, which include lines such as OKABEST™ PAT, OKABEST™ CLX, OKABEST™ CLH, and OKAFLEX™ EM tailored for engineering plastics.
- The company's disclosed range features specific formulations like OKABEST™ CLX 85 (a stabilizer blend for HFFR-PA compounds), OKABEST™ CLX 88 (a multifunctional blend combining antioxidants and phosphorus stabilizers), and OKABEST™ PAT 550B (a combined metal deactivator and antioxidant).

Technical Performance of Antioxidant Grades — scored 78/100

> The rating rests on confirmed technical data showing compliance with industry-standard automotive testing guidelines and independent validation of the antioxidant's thermal efficiency in biopolymers. Based on primary research, the company's comparative performance data can be estimated as robust and exceeding standard market benchmarks, based on its testing results showing better mechanical properties after more than 3,000 hours for polyester compounds and outperforming standard copper-free antioxidants.

SUPPORTING POINTS

- The company reports that its OKABEST PAT 218 heat stabilizer provides long-term temperature resistance exceeding 1,000 hours according to the VW 44045 standard for reinforced polypropylene compounds, while maintaining low volatile and condensable emissions per VDA 275, VDA 277, and VDA 278.
- An independent study published in December 2025 verified that the company's OKABEST CLX84 synthetic antioxidant blend successfully served as a high-performance benchmark for improving the thermal stability of polyhydroxyalkanoates (PHBV) biobased polymers during high-heat processing.

Manufacturing Capacity and Global Footprint — scored 73/100

> Our assessment rests on the company's confirmed singular production base in Germany and its localized warehouse network across the United States and Asia. Based on primary research, the company's manufacturing capacity volumes can be estimated as small to medium scale focused on high-value specialties, based on its operation with 11 to 50 employees at its laboratory and plant in Boenen running on internal solar green electricity.

SUPPORTING POINTS

- The company reports operating a centralized manufacturing plant and laboratory facility in Bönen, Germany, for the production of its specialty polymer stabilizers and antioxidant formulations OKA-Tec GmbH Company Overview.
- The company reports maintaining a global logistics infrastructure with regional warehouses located in the USA and Asia to provide localized product availability and reduce distribution lead times OKA-Tec GmbH Technology Profile.

R&D and Innovation in Antioxidant Chemistry — scored 91/100

> The rating rests on the company's active introduction of specialized, high-performance polymer stabilizer series such as OKABEST and OKAFLEX alongside external research recognition from a German foundation. Based on primary research, the company's annual R&D expenditures can be estimated as highly strategic and recognized by national research bodies, based on its multiple awards from the German Foundation Association including the Innovative by Research accolade.

SUPPORTING POINTS

- OKA-Tec GmbH reports that its research activities in developing long-term heat stabilizers for glass-fiber reinforced polypropylene compounds were awarded the "Innovative by Research 2020/2021" designation by the German Foundation Association.
- In March 2026, the company launched OKABEST CLX 91 and OKABEST CLX 95, which are advanced copper-based stabilizers engineered to provide long-term thermal stability for aliphatic and aromatic polyamides.

Regulatory Compliance and Quality Certifications — scored 88/100

> The rating is based on the company's verified compliance with ISO 9001:2015 and ISO 14001 management standards. Based on primary research, the company's regulatory compliance capability can be estimated as comprehensive and meeting strict automotive and industrial standards, based on its successful DIN EN ISO 24,001 audit certifications alongside compliance with Volkswagen group standards.

SUPPORTING POINTS

- The company announced it successfully transitioned its quality management system to the DIN EN ISO 9001:2015 standard in September 2018.
- The company announced it successfully completed its audit certification according to the ISO 14001 environmental management standard in August 2023.

Y — BUSINESS CAPABILITY

Antioxidant Revenue and Growth — scored 71/100

> We verified that OKA-Tec GmbH produces innovative antioxidant and stabilizer product lines, including OKABEST and OKAFLEX, for engineering plastic applications. Based on primary research, the company's antioxidant additive annual revenue and growth rate can be estimated as growing steadily, based on its expanding specialized product lines like OKABEST and OKAFLEX along with its participation in major international exhibitions.

SUPPORTING POINTS

- Not available

Tier-1 Customer Base and Contracts — scored 65/100

> Based on primary research, the company's direct customer list or long-term supply contracts can be estimated as limited to a few markets, based on its focus on specialized engineering plastics compounders and automotive component manufacturers.

SUPPORTING POINTS

- Not available

Global Sales and Distribution Network — scored 89/100

> We verified that the company maintains an active global presence across Europe, Asia, and the Americas, utilizing external specialty chemical distributors alongside local warehouses in the USA and Asia to manage its global logistics. Based on primary research, the company's quantity, locations, or names of regional sales offices and agency partners can be estimated as broad across major automotive hubs, based on its global presence spanning Germany, France, Spain, Switzerland, Italy, Turkey, India, China, Japan, USA, and Mexico.

SUPPORTING POINTS

- The company reports global coverage across multiple key international markets, showing active presence or operations in Germany, France, Spain, Switzerland, Italy, Turkey, India, China, South Korea, Japan, the USA, and Mexico.
- The company utilizes established external regional distribution networks, including a partnership with Barentz for marketing its high-performance antioxidant products in North America, and historically partnering with Velox for distribution in Turkey, Spain, Portugal, and Israel.

Production Scale and Cost Position — scored 67/100

> We verified that the company maintains an active in-house manufacturing plant and laboratory in Germany for specialized engineering plastic additives. Based on primary research, the company's manufacturing volumes, plant capacities, or pricing metrics can be estimated as strong, in line with leading vendors, based on its state of the art technologies in its Bönen plant and specialized laboratory facilities.

SUPPORTING POINTS

- The company operates a specialized laboratory and manufacturing plant in Bönen, Germany, where it develops and produces its OKABEST and OKAFLEX antioxidant and stabilizer ranges.
- The company reports that its production facility in Bönen has been powered by its own green electricity via an internally installed solar power system since April 2022.

Ownership, Funding, and Strategic Partnerships — scored 77/100

> The rating rests on the company's confirmed commercial distribution and exhibition partnerships within the European specialty chemicals sector. Based on primary research, the company's corporate ownership structure and raw material supply alliances can be estimated as limited to a few markets, based on its independent corporate setup splitting operations between OKA-Tec GmbH and OKA-Tec Vertriebs GmbH.

SUPPORTING POINTS

- OKA-Tec GmbH established a strategic distribution alliance with the specialty materials distributor VELOX in March 2016, authorizing them to market and distribute the company's specialized antioxidants and stabilizers across Spain, Portugal, Turkey, and Israel.
- OKA-Tec GmbH collaborates with Salient Polymers as an exhibition partner to showcase its plastic and compounding additive technologies.

Mayzo, Inc. ● Evolving Players

X 78 Y 77 Overall 78

X — PRODUCT CAPABILITY

Antioxidant Product Portfolio Breadth — scored 83/100

> The company confirms a highly comprehensive portfolio spanning multiple critical antioxidant chemistries and custom blends developed for polymer and elastomer application needs. Based on primary research, the company's custom blend production capacity and formulation details can be estimated as highly flexible and tailored to specific client applications without restrictive volume constraints, based on its 61,000 square foot operations center providing customization on both small and large scales without truckload limitations.

SUPPORTING POINTS

- Mayzo, Inc. reports offering a broad range of antioxidant chemical families for polymer stabilization including phenolic antioxidants, amine antioxidants, phosphites, hydroxylamine systems, thiosynergists (thioethers), and metal deactivators.
- The company reports individual commercial grades within these families, such as 21 specific primary antioxidant grades under its BNX brand, 6 secondary antioxidant grades under its Benefos and BNX brands, and 15 physical stabilizer blends.

Technical Performance of Antioxidant Grades — scored 72/100

> The evaluation rests on the company's detailed product data disclosures detailing specific performance properties—including thermal oxidative protection, gas fading resistance, and extraction resistance—across its BNX antioxidant portfolio for plastics and elastomers. Based on primary research, the company's performance verification and customer validation metrics can be estimated as strong based on application cost and productivity savings, based on its documented custom blending cost reduction and processing advantages.

SUPPORTING POINTS

- The company reports that its BNX 180 primary antioxidant provides low volatility, gas fading resistance, and enhances long-term heat stability when paired synergistically with its thioether secondary antioxidants like BNX 4120.
- The company reports that its liquid antioxidant blend, BNX 1000, provides complete dispersion within adhesives and polymer systems to eliminate traditional powder particle issues and ensure long-term heat stability.

Manufacturing Capacity and Global Footprint — scored 67/100

> We verified that the company's footprint relies on its 61,000-square-foot South Carolina logistics and down-stream processing facility alongside an outsourced global partner network. Based on primary research, the company's annual blending or production capacity metrics can be estimated as high and scalable for diverse batch volumes, based on its 61,000 square foot South Carolina operations facility offering small and large scale solutions.

SUPPORTING POINTS

- The company reports operating the Mayzo Operations Center (MOC) in Walterboro, South Carolina, a facility measuring over 61,000 square feet that handles warehousing, customized packaging, chemical laboratory operations, and additive blending services.
- The company reports utilizing a private-label business model, sourcing its raw antioxidant and chemical additive materials from multiple producer partners across the globe rather than operating its own large-scale chemical synthesis plants.

R&D and Innovation in Antioxidant Chemistry — scored 82/100

> We verified that the company's innovation efforts focus primarily on custom physical form enhancements, physical stabilizer blending, and standard private label packaging rather than novel molecule synthesis or public patent assignments. Based on primary research, the company has not disclosed dedicated R&D spending figures, chemical synthesis patent data, or structured technical research collaborations.

SUPPORTING POINTS

- The company reports the development of BNX 1400, a primary antioxidant designed as an alternative physical form of its standard BNX 1010 chemistry, engineered in 2023 to provide improved free-flow properties and reduced dust for workplace safety.
- The company reports offering its Custom Additive Blends and Custom Solutions programs, under which it develops proprietary antioxidant formulations and physical form enhancements (such as granulation, pelletization, and extrusion) tailored to specific customer application requirements.

Regulatory Compliance and Quality Certifications — scored 88/100

> The rating rests on the company's confirmed ISO 9001:2015 operational facility certification and extensive self-reported FDA clearances for its polymer antioxidant lines. Based on primary research, the company's underlying international regulatory compliance capability can be estimated as rigorous aligned with global manufacturing standards, based on its successful passing of the ACD Responsible Distribution verification and ISO certification.

SUPPORTING POINTS

- The company reports that its Operations Center in Walterboro, South Carolina, achieved ISO 9001:2015 certification for quality management.
- The company reports that multiple products in its antioxidant families—including BNX 1010, BNX 1330, and Benefos 1680—have extensive FDA clearances under specific regulations such as 21 CFR §178.2010 for food-contact polymer applications.

Y — BUSINESS CAPABILITY

Antioxidant Revenue and Growth — scored 82/100

> The assessment rests on the company's self-reported increase in sales and general business growth from its private-label sourcing strategy. Based on primary research, the company's antioxidant sales revenue and growth can be estimated as growing steadily, based on its increase in business and sales due to a flexible and agile sourcing strategy paired up with strong long-term relationships with its suppliers.

SUPPORTING POINTS

- The company reports experiencing an increase in business and credits this increase in sales to a flexible private-label sourcing strategy paired with long-term supplier relationships.

Tier-1 Customer Base and Contracts — scored 68/100

> The evaluation rests on the absence of named Tier-1 direct customers or long-term multi-year supply contracts disclosed by the company within the global antioxidants market. Based on primary research, the company's customer agreements and client list can be estimated as substantial and long term, based on its 35 years of operation supplying global manufacturers across diverse end markets such as plastics, elastomers, and coatings.

SUPPORTING POINTS

- Not available

Global Sales and Distribution Network — scored 84/100

> The evaluation rests on the company's self-reported operations indicating that it acts as a global supplier with international sales coverage and cross-border raw material sourcing. Based on primary research, the company's international channel and distributor network can be estimated as extensive across global markets, based on its global supplier network spanning multiple supplier partners all over the world and its global chemical sales presence.

SUPPORTING POINTS

- The company reports that its chemical additive solutions (including antioxidants and UV stabilizers) are sold on a global scale to international markets throughout the world, supported by a private label supply chain that sources raw materials from producers globally.
- The company maintains its corporate headquarters in Suwanee, Georgia, and a centralized Operations Center in Walterboro, South Carolina, which handles warehousing and distribution services utilizing proximity to the U.S. East Coast ports of Savannah and Charleston.

Production Scale and Cost Position — scored 73/100

> The assessment rests on verified facility infrastructure for blending and repackaging. Based on primary research, the company's antioxidant production and manufacturing costs can be estimated as efficient and highly scalable, based on its 61,000 square foot operations center providing custom blending, repackaging, and physical form enhancement near major shipping ports.

SUPPORTING POINTS

- The company operates a 61,000+ square foot Operations Center in Walterboro, South Carolina, which handles warehousing, chemical additive blending, repackaging, and custom physical form adjustments.
- The company reports offering custom blending, packaging, and product form customization with flexible low minimum quantity requirements for small and large buyers.

Ownership, Funding, and Strategic Partnerships — scored 77/100

> The rating rests on the company's long-term private ownership structure and its completed acquisition of an additives distributor to expand its supply network. Based on primary research, the company's institutional funding and joint ventures can be estimated as highly structured around strategic industry alliances, based on its exclusive rights to manage sales segments for major global chemical groups like basf into specific adhesive markets.

SUPPORTING POINTS

- The company reports that it operates as a privately held specialty chemical supplier founded in 1986 by Ben Milazzo.
- Mayzo completed the acquisition of specialty chemical distributor Bio Accutech Inc. on October 9, 2020, to extend its global supply chain for additives.

ICG Specialty Chemicals FZCO OMNIQUAN ● Evolving Players X 77 Y 77 Overall 77

X — PRODUCT CAPABILITY

Antioxidant Product Portfolio Breadth — scored 83/100

> The rating rests on the company's active marketing of multiple key antioxidant chemistries under its proprietary brands, including aromatic amines, hindered phenolics, organophosphites, and thioesters. Based on primary research, the company's quantitative count of commercial grades can be estimated as comprehensive across primary and secondary stabilizer types, based on its extensive product catalog including primary hindered phenolics, secondary phosphites, and specialized metal deactivators.

SUPPORTING POINTS

- The company reports offering liquid aromatic amine antioxidants, such as OMNIQUAN AN 5057, for polyol, elastomer, and adhesive applications.
- The company reports supplying primary hindered phenolic antioxidants (e.g., OMNIQUAN AN 245, AN 1024 MD, AN 1135, AN 1425, AN 1520, AN 1726, and AN 3114), secondary organophosphite processing stabilizers (e.g., AN 168 and AN 626), and secondary sulfur-containing thioester antioxidants (e.g., OMNIQUAN AN 800).

Technical Performance of Antioxidant Grades — scored 72/100

> The rating rests on the company's self-reported technical specifications and performance advantages for its specialized hindered phenolic, sulfur, and aminic antioxidant grades targeted at polymer and elastomer compounding. Based on primary research, the company's test logs can be estimated as strong, in line with leading vendors, based on its published data sheets specifying melting points, purity thresholds, and synergistic polymer processing benefits.

SUPPORTING POINTS

- The company reports that its Omniquan AN 1425 antioxidant grade functions as a high molecular weight antioxidant and multifunctional stabilizer for plastics, elastomers, and rubber, offering specific technical properties including color stability with blue-shift properties, high extraction resistance, and low volatility.
- The company reports that its Omniquan AN 1330 grade provides specialized technical performance for electrically demanding applications like wire and cable, delivering key properties such as low volatility, high resistance to water extraction, and specific dielectric and electrical insulation performance.

Manufacturing Capacity and Global Footprint — scored 67/100

> We verified that the company operates as a specialty chemical supplier and distributor originally set up as a regional branch office in Dubai, utilizing contract or joint venture arrangements in China for its proprietary brands like Omnistab and Omniquan. Based on primary research, the company's total annual production capacity can be estimated as broad across international supply corridors, based on its strategic joint venture with Deltachem Qingdao for sourcing, toll manufacturing, and corporate product line distribution.

SUPPORTING POINTS

- The company reported a 20 million yuan investment in 2016 for the production of light stabilizers and antioxidants through its historical joint venture / footprint arrangement with Deltachem Qingdao in China, which focused on sourcing, toll manufacturing, and production of its Omnistab range.

R&D and Innovation in Antioxidant Chemistry — scored 90/100

> The rating rests on the company's verified commercial partnerships dating back to 2016 for the production and distribution of its established Omnistab antioxidant range. Based on primary research, the company's recent R&D investments can be estimated as focused on functional application expansions, based on its recent product additions targeting r-PET performance enhancement, wire resins, and polyurethane stabilization.

SUPPORTING POINTS

- The company expanded its portfolio in July 2016 by producing light stabilizers and antioxidants under its proprietary Omnistab brand, collaborating with Deltachem Co., Ltd. and Tangshan Longquan Chemical Auxiliary Co., Ltd. in China.
- The company reports marketing and distribution of standard chemical antioxidant formulas, including OMNIQUAN AN BHT, OMNISTAB AN 168, OMNIQUAN AN 1520, and OMNIQUAN AN 1726, for thermo-oxidative protection and processing stability in plastic and elastomer applications.

Regulatory Compliance and Quality Certifications — scored 74/100

> The rating rests on the company's self-reported claim regarding the regulatory suitability of its OMNIQUAN AN 800 product for specific food-contact and cosmetic applications. Based on primary research, the company's quality certifications can be estimated as rigorous for major global entry requirements, based on its published application listings compliance guidelines for food-contact packaging materials and cosmetic polymer substrates.

SUPPORTING POINTS

- The company reports that its secondary sulfur-containing antioxidant OMNIQUAN AN 800 is suitable for selected cosmetic and food-contact applications, subject to applicable regulatory requirements.

Y — BUSINESS CAPABILITY

Antioxidant Revenue and Growth — scored 71/100

> Based on primary research, the company's annual revenue and growth rate can be estimated as growing steadily, based on its over 16 years of expertise and globally trusted brands omnistab and omniquan.

SUPPORTING POINTS

- Not available

Tier-1 Customer Base and Contracts — scored 65/100

> Based on primary research, the company's customer list or long-term supply agreements can be estimated as broad across the major polymer sectors, based on its serving of diverse industries including masterbatch, plastics, resins, compounding, inks, and coatings across the middle east and asia.

SUPPORTING POINTS

- Not available

Global Sales and Distribution Network — scored 91/100

> The rating rests on the company's integrated sales and distribution reach through its parent group, Partners in Chemicals, which supports active branch operations and localized distribution networks covering Europe, the Americas, Asia, the Middle East, and Africa. Based on primary research, the company's count of distribution agents can be estimated as centralized from regional operational bases, based on its remote day-to-day coordination by international operations managers managing global suppliers and clients from key hubs.

SUPPORTING POINTS

- The company operates as a part of the Partners in Chemicals group, established in 2014 as the Dubai branch office (ICG Specialty Chemicals FZCO) to handle sales activities across the Middle East, Africa, and India.
- Through its parent group, Partners in Chemicals, the sales and distribution network covers global markets via dedicated local partners and branch offices across Europe (Utrecht headquarters), the Americas (Chicago branch office established in 2009), the Middle East, and China (via the Deltachem Qingdao joint venture established in 2012).

Production Scale and Cost Position — scored 81/100

> The evaluation rests on the company's self-reported capability to supply antioxidants in bulk commercial quantities and large-volume configurations. Based on primary research, the company's production volumes or cost-efficiency metrics can be estimated as optimized for large industrial demands, based on its supply of a deep portfolio of high-performance phenolic primary and organophosphite secondary antioxidants with local stocking in its dubai warehouse.

SUPPORTING POINTS

- The company reports providing bulk commercial quantities, large-volume supply capacity, and reliable production capacity for its primary and secondary antioxidant ranges, including its Omniquan series.
- Through its historical joint venture with Deltachem (Qingdao) Co. established in 2009, the group established sourcing, manufacturing, and production capabilities for its chemical product lines.

Ownership, Funding, and Strategic Partnerships — scored 75/100

> The rating rests on its history as a former regional branch of the Partners in Chemicals group followed by a complete management buy-out in 2019 by its regional director. Based on primary research, the company's external funding rounds or institutional backing can be estimated as self-sustaining through independent ownership, based on its transition to an independent management buy-out structure where director Ishfaq Ahmed took over all company shares.

SUPPORTING POINTS

- The company was originally established as a Dubai branch office of the Dutch group Partners in Chemicals in 2012 to handle sales operations across the Middle East, Africa, and India.
- In 2022, a management buy-out was completed by the Dubai branch's director, Mr. Ishfaq Ahmed, who took over all shares to operate the business independently under ICG Specialty Chemicals.

Transform Master Sdn Bhd (subsidiary of Luxchem Corporation Berhad) TRANSNOX ● Evolving Players

X 75 Y 79 Overall 77

X — PRODUCT CAPABILITY

Antioxidant Product Portfolio Breadth — scored 73/100

> The assessment rests on the confirmation that the company manufactures a dedicated antioxidant product line under the TRANSNOX AO50 brand and general AO emulsions for the rubber and latex industries. Based on primary research, the company's underlying chemical chemistries and commercial grades can be estimated as focused and specialized, based on its TRANSNOX AO50 antioxidant product and focused portfolio for rubber latex industries.

SUPPORTING POINTS

- The company reports offering a portfolio of latex compounding and processing chemicals that includes the specific antioxidant dispersion grade TRANSNOX AO50.
- Industry product directories confirm that the company manufactures latex chemical dispersions and composite materials, including an AO Emulsion (antioxidant emulsion) product line.

Technical Performance of Antioxidant Grades — scored 68/100

> We verified that the company manufactures a dedicated antioxidant grade (TRANSNOX AO50) for the rubber latex industry and handles commercial quantities of bulk antioxidants. Based on primary research, the company's performance test figures and comparative aging data can be estimated as verified and compliant, based on its established reputation in design and development supported by well-equipped testing laboratories.

SUPPORTING POINTS

- The company reports offering TRANSNOX AO50, a latex-specific chemical dispersion antioxidant, primarily formulated for applications including medical gloves, latex toys, and foam.
- Bill of lading records indicate that Transform Master Sdn Bhd imported 20 metric tons of Antioxidant KY616 to Malaysia as of June 2026.

Manufacturing Capacity and Global Footprint — scored 65/100

> We verified that the company manufactures rubber latex chemical dispersions, including antioxidants, at its single production site in Perak, Malaysia. Based on primary research, the company's production capacity and manufacturing footprints can be estimated as centralized within regional port zones, based on its manufacturing operations located solely at Lumut Port Industrial Park in Sitiawan.

SUPPORTING POINTS

- The company reports operating its manufacturing facilities at the Lumut Port Industrial Park in Sitiawan, Perak, Malaysia, where it manufactures latex compounding and processing chemicals including its TRANSNOX AO50 antioxidant.

R&D and Innovation in Antioxidant Chemistry — scored 87/100

> The rating rests on the company's self-reported laboratory and process development capabilities for its latex antioxidant dispersions. Based on primary research, the company's technical collaborations and R&D financial investments can be estimated as continuous and specialized, based on its strong process development capability and integration within the parent Luxchem Group structure.

SUPPORTING POINTS

- The company reports having process development capability and laboratories to design, develop, and manufacture latex compounding chemicals, including its TRANSNOX AO50 antioxidant dispersion.

Regulatory Compliance and Quality Certifications — scored 84/100

> We verified that the company holds valid ISO 9001:2015 quality management and ISO 14001:2015 environmental certifications for its manufacturing operations. Based on primary research, the company's chemical regulatory compliance and food-contact approvals can be estimated as compliant with major global frameworks, based on its active compliance with international standards and manufacturing alignment with medical glove applications.

SUPPORTING POINTS

- The company reports that its manufacturing facility is certified with the ISO 9001:2015 Quality Management System and the ISO 14001:2015 Environmental Management System.
- The company manufactures rubber and latex chemical dispersions, including its TRANSNOX AO50 antioxidant formulation.

Y — BUSINESS CAPABILITY

Antioxidant Revenue and Growth — scored 72/100

> We verified that the company manufactures antioxidant chemical dispersions for the latex and rubber industries, but current separate revenue figures for this specific product line are absent. Based on primary research, the company's annual antioxidant revenue and growth rate can be estimated as growing steadily, based on its position as an industrial chemical manufacturer whose parent reported a net sales revenue increase of over 47%.

SUPPORTING POINTS

- The company reports offering TRANSNOX AO50, an antioxidant dispersion, as part of its latex compounding and processing chemical formulations.
- Before its acquisition by Luxchem Corporation Berhad, the company had an annual revenue of MYR 30.9 million for the year ended December 31, 2015.

Tier-1 Customer Base and Contracts — scored 68/100

> Based on primary research, the company's prominent rubber and latex client identities can be estimated as concentrated within the rubber latex sector, based on its focused design, development, and manufacturing of compounding chemicals primarily for medical gloves, condoms, and catheters.

SUPPORTING POINTS

- Not available

Global Sales and Distribution Network — scored 78/100

> The assessment rests on the verified location of the company's production facilities and its established shipping activities to regional international markets through its parent corporation. Based on primary research, the company's international distributor and sales network extent can be estimated as integrated with parent international distribution channels, based on its parent group distribution network that exports over 400 types of chemicals across 13 countries including Singapore, Indonesia, and Vietnam.

SUPPORTING POINTS

- The company is a Malaysian manufacturer based at Lumut Port Industrial Park in Kampung Acheh, Sitiawan, Perak, and operates as a subsidiary of Luxchem Corporation Berhad.
- Shipment data indicates that the company exports chemical products, including rubber antioxidants, internationally to locations such as Vietnam.

Production Scale and Cost Position — scored 88/100

> The assessment rests on the company’s documented monthly plant capacity of 1,500 metric tons and its high utilization rates under its parent company, Luxchem Corporation Berhad. Based on primary research, the company's antioxidant line manufacturing cost breakdown can be estimated as competitive and optimized for volume production, based on its production facilities located at the Lumut Port Industrial Park supported by specialized process development capabilities and laboratories.

SUPPORTING POINTS

- Transform Master Sdn Bhd recorded a total monthly production capacity of approximately 1,500 metric tons (equivalent to an annual capacity of up to 18,000 metric tons), maintaining a 93% factory utilization rate during FY2024 and reaching 75% capacity utilization in Q1 2025.
- The company manufactures chemical dispersions for the latex and rubber industries, including its proprietary TRANSNOX AO50 antioxidant.

Ownership, Funding, and Strategic Partnerships — scored 90/100

> The rating is based on the company's clear corporate backing as a wholly-owned subsidiary of the established chemical group Luxchem Corporation Berhad and its verified manufacture of rubber latex antioxidants under the TRANSNOX brand. Based on primary research, the company's funding rounds and strategic partnerships can be estimated as stable and well-supported, based on its corporate structure backed entirely by its parent entity Luxchem Corporation Berhad with over 40 years of industrial chemical experience.

SUPPORTING POINTS

- Transform Master Sdn Bhd was acquired on 29 April 2016 and operates as a wholly-owned subsidiary of Luxchem Corporation Berhad, a public investment holding company listed on Bursa Malaysia.
- The company manufactures a specialized range of latex compounding and processing chemicals for the rubber latex industry, which includes its own proprietary antioxidant product line branded as TRANSNOX AO50.

Coraplus Pte. Ltd. XONOX ● Evolving Players X 75 Y 77 Overall 76

X — PRODUCT CAPABILITY

Antioxidant Product Portfolio Breadth — scored 82/100

> The rating rests on the company's confirmed supply of primary phenolic and secondary phosphite chemistries under its XONOX product portfolio. Based on primary research, the company's breadth of grades or technical specifications can be estimated as comprehensive across primary and secondary applications, based on its primary amine and phenolic classes and secondary phosphite and sulfide classes.

SUPPORTING POINTS

- The company reports through its product documentation that its XONOX brand includes XONOX 168, a hydrolytically stable phosphite processing stabilizer acting as a secondary antioxidant.
- The company reports through its product documentation that its XONOX line features XONOX 1010, which acts as a primary phenolic antioxidant used in combination with secondary stabilizers to provide synergistic long-term thermal stability.

Technical Performance of Antioxidant Grades — scored 66/100

> The rating rests on Coraplus Pte. Ltd.'s high-level product descriptions of its XONOX antioxidant portfolio, which outlines general primary and secondary chemical classes such as amines, phenolics, phosphites, and thioesters. Based on primary research, the company's validation performance metrics can be estimated as excellent thermal stability and long-term thermal stabilization, based on its XONOX 1010V high extraction resistance and low volatility.

SUPPORTING POINTS

- The company reports that its XONOX 168 product grade is a hydrolytically stable phosphite processing stabilizer acting as a secondary antioxidant that reacts with available hydroperoxides formed by autoxidation to prevent process-induced degradation and extend primary antioxidant performance.

Manufacturing Capacity and Global Footprint — scored 65/100

> We verified that the company maintains its primary corporate and R&D presence in Singapore and utilizes international distribution networks. Based on primary research, the company's physical manufacturing plant location or capacity can be estimated as structured for global supply and additive package development, based on its ISO 9001:2015 certification and international distribution networks.

SUPPORTING POINTS

- The company reports operating its Research and Development Headquarters from the Manulife Tower in Singapore, while maintaining an official representative network via Kemipex FZE in Dubai, UAE to cover the Middle East, Africa, and Europe.

R&D and Innovation in Antioxidant Chemistry — scored 82/100

> The assessment is based on the company's self-reported marketing of its XONOX antioxidant portfolio covering common primary and secondary chemical structures for polymer stabilization. Based on primary research, the company's technical collaborations or molecule launches can be estimated as focused on polymer lifetime extension and specialized additive packages, based on its core LUBIMAX, CARBIMAX, MAXXPOL, EPOCURE, and XONOX portfolios.

SUPPORTING POINTS

- The company reports that its XONOX brand offers primary (aromatic amines and sterically hindered phenols), secondary (phosphites, phosphonites, and sulfides), and customized blend antioxidant formulations to enhance the thermal and processing stability of polymer and plastic applications.

Regulatory Compliance and Quality Certifications — scored 82/100

> We verified that the company's assessment relies solely on its self-reported ISO 9001:2015 quality management certification. Based on primary research, the company's regulatory approvals or compliance certificates can be estimated as highly compliant with international consumer safety mandates, based on its FDA compliance for XONOX 1010V and XONOX L 57 food-contact applications.

SUPPORTING POINTS

- Coraplus Pte. Ltd. has achieved ISO 9001:2015 certification for its quality management system as a chemical manufacturer and formulation developer, as reported on the Coraplus Official Website.

Y — BUSINESS CAPABILITY

Antioxidant Revenue and Growth — scored 83/100

> We verified financial reporting ranges for the company's regional subsidiary. Based on primary research, the company's annual revenue and growth rates can be estimated as strong, in line with growing demand, based on its position as a live private company registered since 2,017 and offering wide chemical portfolios.

SUPPORTING POINTS

- The company reports its unlisted subsidiary, Coraplus India Private Limited, generated operating revenues in the range of INR 1 cr - 100 cr for the financial year ending on 31 March, 2025.

Tier-1 Customer Base and Contracts — scored 65/100

> We verified that the company offers its polymer antioxidants globally under the XONOX brand. Based on primary research, the company's enterprise customer base can be estimated as active across diverse polymer and industrial applications, based on its specialized formulation capabilities designed to meet client-specific requirements and industrial buyer specifications.

SUPPORTING POINTS

- Not available

Global Sales and Distribution Network — scored 87/100

> The rating rests on confirmed regional office presence in Singapore, India, and the UAE, alongside an authorized distributorship agreement with Kemipex FZE to span Europe, the Middle East, and Africa. Based on primary research, the company's distribution network and regional channel partners can be estimated as growing globally from regional chemical hubs, based on its strategic placement in singapore alongside multinational offices located in india and dubai.

SUPPORTING POINTS

- Coraplus reports corporate office locations in Singapore (Marina Bay Financial Centre), Dubai (Capricorn Tower, Sheikh Zayed Road), and India (MIDC, Wagle Industrial Estate, Thane West) to anchor its international sales and distribution network for its industrial additive lines, including its XONOX polymer antioxidants.
- Technical documentation reveals that Coraplus operates with Kemipex FZE, located in Dubai, UAE, as its official representative covering markets across the Middle East, Africa, and Europe.

Production Scale and Cost Position — scored 73/100

> We verified that the company manufactures a dedicated line of polymer antioxidants under the XONOX™ brand. Based on primary research, the company's manufacturing capacity and volumes can be estimated as focused on specialized formulations rather than scale, based on its technical depth and formulation-based model designed to exceed specific customer requirements.

SUPPORTING POINTS

- The company reports marketing its line of high-performance polymer antioxidants under the brand name XONOX™, offering primary, secondary, and custom blends of amines, phenolics, phosphites, and thioesters to improve the processing and long-term thermal stability of plastics.

Ownership, Funding, and Strategic Partnerships — scored 77/100

> We verified that the company operates as a privately held subsidiary under Vestcorp group holdings, giving it structured organizational backing. Based on primary research, the company's funding amounts or joint ventures can be estimated as steady and self-sustained by corporate equity, based on its initial private limited registration with standard paid up capital and long-term operating status.

SUPPORTING POINTS

- Vestcorp corporate documentation explicitly identifies Coraplus Pte. Ltd. Singapore as a subsidiary within the Vestcorp group holdings.
- The company reports that it leverages strategic collaborations with suppliers, research institutions, and industry experts to integrate advanced tech into its customized chemical solutions.

HECOPLAST GmbH HECONOX ● Evolving Players X 77 Y 65 Overall 71

X — PRODUCT CAPABILITY

Antioxidant Product Portfolio Breadth — scored 79/100

> The rating rests on the company's disclosed line of six standard granular antioxidant masterbatches under its proprietary additive brand. Based on primary research, the company's chemical classifications can be estimated as primary acting radical scavengers and secondary peroxide decomposers, based on its primary antioxidants acting as radical scavengers and secondary antioxidants acting as peroxide decomposers.

SUPPORTING POINTS

- The company reports offering its HECO®NOX range of antioxidant additives for the plastics industry, which includes primary radical scavengers and secondary peroxide decomposers across six specific granular grades: 603 PP, 604 PS, 605 PA, 620 PA, 625 PE, and 650 CYC.
- Lists 8 antioxidant products under the HECOPLAST brand within its polymer additives platform directory.

Technical Performance of Antioxidant Grades — scored 65/100

> Our assessment rests on the company's self-reported functional properties for its HECO®NOX and HECO® STAB additive grades regarding thermal stability and aging protection. Based on primary research, the company's performance validation can be estimated as highly effective for preventing thermal aging and polymer degradation, based on its non-discoloring formulations that serve as process and long-term thermo-stabilizers.

SUPPORTING POINTS

- The company reports that its HECO®NOX line of antioxidants—including primary antioxidants like HECO®NOX 604 PS and additive mixtures like HECO®NOX 625 PE and HECO®NOX 605 PA—renders plastics thermally stable, extends their service life, and maintains flexibility under heat-related decomposition.
- The company reports that its HECO® STAB 376 PP grade operates as a combined light stabilizer and antioxidant that provides a thermo-stabilizing effect and prevents polypropylene polymer degradation initiated by UV radiation.

Manufacturing Capacity and Global Footprint — scored 67/100

> We verified that the company maintains a centralized development and production site in Germany for its antioxidant masterbatch line. Based on primary research, the company's manufacturing volume can be estimated as growing following production facility expansion, based on its recent relocation to an expanded headquarters to increase production capacity.

SUPPORTING POINTS

- The company reports relocating its headquarters to a new facility on Zollhausstrasse in Iserlohn, Germany, in August 2024 to provide expanded space to increase production capacity.
- The company reports producing a line of polymer antioxidant masterbatches under the HECO-NOX brand, operating primarily out of its facility in Germany with 11 to 50 employees.

R&D and Innovation in Antioxidant Chemistry — scored 79/100

> The assessment rests on the company's general self-reported commitment to developing and refining its polymer additive formulations. Based on primary research, the company's technical collaborations can be estimated as strong custom development based on specific polymer issues, based on its custom research and development focused on international cosmetics, pharmaceutical, and packaging customers.

SUPPORTING POINTS

- The company reports that it continuously conducts research and development to create new products and improve its established plastic additives portfolio.

Regulatory Compliance and Quality Certifications — scored 93/100

> Our assessment rests on the company's verified third-party quality management system certification and specific European food-contact regulatory alignment for its polymer additives. Based on primary research, the company's regulatory approvals can be estimated as broad across international compliance frameworks, based on its compliance certifications spanning European Union food contact standards, US FDA approvals, and Pharmacopeia criteria.

SUPPORTING POINTS

- HECOPLAST GmbH holds a DIN EN ISO 9001:2015 quality management system certification issued by TÜV NORD CERT GmbH for the development, technical consulting, production, and distribution of plastic and foam additives, which is valid from September 27, 2025, until September 26, 2028.
- The company offers antioxidant products within its HECO® NOX line that comply with Regulation (EU) No. 10/2011 concerning plastic materials and articles intended to come into contact with food.

Y — BUSINESS CAPABILITY

Antioxidant Revenue and Growth — scored 65/100

> Based on primary research, the company's annual revenue and growth rate can be estimated as growing steadily, based on its growing requirements and recent relocation to a larger headquarters to expand production capacity.

SUPPORTING POINTS

- Not available

Tier-1 Customer Base and Contracts — scored 65/100

> Our evaluation rests on the absence of documented contract announcements or verified commercial partnerships connecting the company to major polymer, tire, or plastics manufacturers. Based on primary research, the company's named tier-1 customers can be estimated as broad across target industries, based on its international operating customers from the cosmetics, packaging, closure, pharmaceutical, and household goods sectors.

SUPPORTING POINTS

- Not available

Global Sales and Distribution Network — scored 65/100

> The evaluation rests on the confirmed finding that HECOPLAST GmbH handles its global customer base from its centralized corporate and production headquarters in Germany without a distributed physical footprint. Based on primary research, the company's international distributor network can be estimated as limited to a few markets, based on its reliance on global trade fairs like FAKUMA, KUTENO, and KPA Ulm to reach buyers.

SUPPORTING POINTS

- The company reports serving internationally operating customers from the household goods, packaging, closure, cosmetics, and pharmaceutical industries worldwide.
- We verified that the company operates out of a single centralized headquarters and production location in Iserlohn, Germany, and engages in direct marketing within Europe by exhibiting at regional trade fairs like FAKUMA and KUTENO.

Production Scale and Cost Position — scored 67/100

> We verified that the company expanded its facility footprint to accommodate growing operational demands. Based on primary research, the company's antioxidant production volumes can be estimated as high for custom batches, based on its comprehensive portfolio of HECO-NOX masterbatch granules tailored for diverse polymers.

SUPPORTING POINTS

- The company reports that it relocated its headquarters to new premises providing a larger area to expand its production capacity.

Ownership, Funding, and Strategic Partnerships — scored 65/100

> The rating rests on the company's verified identity as an independent, family-managed German additive developer, alongside its recent 2026 global e-distribution partnership with to market its HECO®NOX plastic antioxidants. Based on primary research, the company's corporate financial backing can be estimated as limited to family ownership, based on its operational history as a medium-sized family business managed by its founders.

SUPPORTING POINTS

- HECOPLAST GmbH is a medium-sized German family business founded in 1996 by Wolfgang Henzler and currently managed by Ina Henzler.
- The company reports a strategic distribution collaboration with initiated in 2026 to make its innovative polymer additives portfolio, including the HECO®NOX antioxidant series, globally accessible.

Parameter Definitions

X — Product Capability

1. **Antioxidant Product Portfolio Breadth**
   The range of antioxidant chemistries offered, including aromatic amines, hindered phenolics, phosphites, and thioesters, and the number of specific grades within each. Strength is shown by a broad portfolio covering multiple polymer stabilization needs.
2. **Technical Performance of Antioxidant Grades**
   The effectiveness of antioxidants in key properties such as thermal stability, color retention, and long-term protection, often measured by industry-standard tests. Strength is evidenced by superior performance data and customer validation.
3. **Manufacturing Capacity and Global Footprint**
   The total production capacity for antioxidant additives and the geographic spread of manufacturing plants. Strength is indicated by large-scale capacity and multiple sites serving key regions.
4. **R&D and Innovation in Antioxidant Chemistry**
   Investment in research and development to create new antioxidant molecules, formulations, and sustainable solutions. Strength is shown by patents, new product launches, and technical collaborations.
5. **Regulatory Compliance and Quality Certifications**
   Compliance with global chemical regulations (e.g., REACH, FDA) and quality management certifications (e.g., ISO 9001). Strength is demonstrated by a comprehensive set of approvals and certifications for food-contact and other sensitive applications.

Y — Business Capability

1. **Antioxidant Revenue and Growth**
   The annual revenue generated specifically from antioxidant additive sales and its growth rate over recent years. Strength is evidenced by high revenue figures and consistent positive growth.
2. **Tier-1 Customer Base and Contracts**
   The number and prominence of direct customers, particularly large tire, rubber, and plastics manufacturers, and the existence of long-term supply agreements. Strength is shown by named Tier-1 customers and multi-year contracts.
3. **Global Sales and Distribution Network**
   The extent of sales offices, distributors, and agents covering key markets worldwide. Strength is indicated by a presence in major regions and a broad network of channel partners.
4. **Production Scale and Cost Position**
   The volume of antioxidants produced and the company's ability to compete on price through economies of scale and efficient processes. Strength is shown by high production volumes and competitive pricing.
5. **Ownership, Funding, and Strategic Partnerships**
   The company's ownership structure, access to capital, and alliances with raw material suppliers or technology providers. Strength is evidenced by backing from large chemical groups, recent investments, and strategic joint ventures.

Research Methodology

This quadrant was compiled independently through structured secondary research, supplemented by primary research where available. It is not based on a survey or on vendor self-submission. The steps below describe how companies were identified, screened, evidenced and scored.

1

Market Definition

The market scope and the type of company that qualifies were defined first. Five parameters were then set for each axis, each with a written definition: the X axis measures core capability (products, services or technology) and the Y axis measures business capability.

→

2

Secondary and Primary Research

Secondary research covered company websites, filings, trade directories, industry publications and news. Primary research drew on B2B marketplace listings, where available, to confirm company offerings. Coverage was extended region by region until no new qualifying companies emerged.

→

3

Screening

Each candidate was screened against the market's inclusion criteria and required commercial role. Distributors, resellers, consultancies, contract manufacturers and media names were excluded, and duplicates consolidated. Companies whose main business is an adjacent segment are listed but not placed on the chart.

→

4

Evidence Compilation

Every screened company was assessed on all ten parameters; for chart companies, evidence was recorded parameter by parameter. Claims resting only on a company's own materials are reported as company-reported, partner technology is credited to the partnership, and employee-review sites are not used as customer evidence.

→

5

Scoring

Each parameter is rated 0–100 against its written definition, based on the evidence available. Ratings are analyst assessments under this methodology, not independently audited figures. X and Y are the simple average of the five parameter ratings on each axis; overall is the average of X and Y.

→

6

Normalization & Selection

X and Y scores are rescaled to a 65–100 range across all companies assessed, preserving their rank order. The chart shows an equal number of companies from each of the four quadrants, with boundaries set at the median. All other companies are listed in the table below the chart.

Scoring inputs: each axis is built from five market-specific parameters, defined in the Parameter Definitions section. All parameters are equally weighted.